

# Perspectives and Energy Applications of Magnetocaloric, Pyromagnetic, Electrocaloric, and Pyroelectric Materials

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This perspective provides an overview of the state of research and innovation in the areas of magnetocaloric and pyromagnetic materials, and electrocaloric and pyroelectric materials, including the overlapping sub-areas of multicaloric and multipyro materials that can operate simultaneously under the application of magnetic and electric fields. These materials are critically examined for their potential to revolutionize cooling, heating, and energy-harvesting applications. This perspective first summarizes the state-of-the-art advancements and highlights recent significant developments. Then, it is identified and discussed that the prevailing challenges hindering the widespread adoption of technologies based on these materials. In this context, after consulting with members of the caloric and pyro communities, a technology roadmap is outlined to guide research efforts in overcoming current barriers to applications, with the goal of achieving impactful results by 2040. This roadmap emphasizes the importance of focusing on under-researched materials, novel devices, and application spaces, paving the way for interdisciplinary efforts that can lead to significant reductions in carbon dioxide emissions.

accounting for approximately the other half. These everyday applications contribute significantly to air pollution and to over 40% of global energy-related CO<sub>2</sub> emissions.<sup>[1]</sup> Moreover, it is expected that demand for refrigeration and air conditioning, which currently represent ≈17% of global electricity consumption,<sup>[2]</sup> will triple by 2050;<sup>[3]</sup> high growth is also projected for heat pumps.<sup>[4]</sup> Large amounts of waste heat are continually released into the environment. According to Firth et al.<sup>[5]</sup> Nearly 50% of the annual global energy consumption will result in waste heat by 2030. In another study, the relative contributions of global waste heat are broken down by relevant temperature levels: 63% below 100 °C, 16% between 100 and 300 °C, and 21% above 300 °C.<sup>[6]</sup>

The majority of global CO<sub>2</sub> emission reductions can be achieved via energy-efficient heating and cooling, the use

## 1. Introduction

Nearly 50% of global end-energy consumption is attributed to heating and cooling, with industrial processes accounting for approximately half, and residential and commercial buildings

of environmentally friendly refrigerants, heat recovery, waste heat harvesting, and the implementation of efficient thermal management protocols, which can significantly improve the performance of these technologies. The coupling of sectors using power-to-heat approaches can maximize energy efficiency by circumventing direct (e.g., resistive) electrical energy conversion or fuel combustion methods, which demonstrate notably low second-law efficiency when generating low- or medium-temperature heat.<sup>[7]</sup>

The most widely used technology for active heating and cooling, known for its energy efficiency, is vapor compression. However, this established technology faces challenges related to the environmental footprint, toxicity, flammability, and explosion hazards associated with refrigerants.<sup>[8]</sup> Similar problems affect organic Rankine cycles (ORCs), which are the most energy-efficient methods for producing heat-to-power via low- to medium-temperature heat sources.<sup>[9]</sup> In general, vapor compression and ORCs suffer from low exergy efficiency in small-scale devices. While vapor compression refrigeration and heat pumping are limited to operation between −100<sup>[10]</sup> and 200 °C with few exceptions,<sup>[11]</sup> ORCs enable the use of heat sources from 100<sup>[12]</sup> to 450 °C.<sup>[13]</sup> Sorption technologies, such as absorption and adsorption refrigeration and heat pumping, are particularly crucial for the utilization of waste heat or renewable heat sources.<sup>[14–16]</sup> They are generally less efficient than vapor compression and have a significantly narrower temperature range from ≈−40 °C for refrigeration<sup>[17]</sup> to ≈160 °C for heat pumping. Another

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DOI: 10.1002/aenm.202401739

heat-to-power technology is the Kalina cycle,<sup>[18]</sup> which requires a more complex design (as well as higher cost) and operation than an ORC.<sup>[19]</sup> It also poses risks related to the use of ammonia as a working substance. The only solid-state technology available on the market is thermoelectric heating and cooling based on the Peltier effect;<sup>[20]</sup> however, it yields a very low exergy efficiency, which also applies to its counterpart for power generation based on the Seebeck effect.<sup>[21]</sup> Special attention must be paid to improving cryogenic gas-cycle refrigeration, which, for example, has an exergy efficiency between 5% and 25% when cooling to 80 K.<sup>[22]</sup>

Caloric cooling, heating, and pyro energy harvesting based on ferroic materials are promising solid-state technologies that utilize thermodynamic cycles and functional materials as refrigerants. These technologies are compelling alternatives to current market options. Their main advantage lies in the near-reversibility of the caloric effects, in contrast to the irreversible processes of traditional gas compression or expansion. This potential reversibility could lead to significantly higher energy efficiency in refrigeration, heat pumping, and power generation. Moreover, solid-state refrigerants are leak-free and recyclable, further increasing their attractiveness.

This perspective discusses the state-of-the-art advancements, challenges, and future prospects for heating, cooling, and thermal energy harvesting, with a focus on the use of field-driven i-caloric and pyro-i phenomena, for which applications do not necessarily demand moving parts, such as magnetocaloric, pyromagnetic, electrocaloric, and pyroelectric materials. This perspective also includes a technology development roadmap for 2040, focusing on materials and devices for heating, cooling, and thermal energy harvesting, with particular attention to under-researched materials and applications.

## 2. Thermodynamics of Pyro and Caloric Materials

The following section describes the thermodynamics of magnetocaloric/pyromagnetic materials, and electrocaloric/pyroelectric materials. Several reviews and books have separately described the thermodynamics of magnetocaloric, pyromagnetic, electrocaloric, and pyroelectric materials separately (for example, refs.).<sup>[23–28]</sup> Therefore, we briefly discuss the thermodynamics section. We used a unified thermodynamic notation to simultaneously describe both sets of materials.

We began by considering that the thermal, mechanical, and polarization behavior of a ferroic homogeneous material is described by temperature  $T$ , entropy  $S$ , field  $\Phi$ , and induction  $I$ . Induction  $I$  is defined as  $I = \kappa\Phi + \Pi$ , where polarization  $\Pi$  is the density of dipole moments in the material, and  $\kappa$  is a constant. Induction  $I$ , also known as displacement, differs from field  $\Phi$  because it accounts for the effect of polarization  $\Pi$  in materials, such that in vacuum  $I = \kappa\Phi$ ,  $\kappa$  is the permeability of free space  $\mu_0$  for magnetocaloric/pyromagnetic materials, and the permittivity of free space  $\epsilon_0$  for electrocaloric/pyroelectric materials.

The definitions of the i-caloric and pyro-i effects must be noted. Caloric effects refer to the temperature change in a material due to the application or removal of a driving magnetic field  $\Phi$ . As indicated in **Figure 1a**, depending on the type of caloric effect, we focused on the magnetic or electric fields for the magnetocaloric and electro-caloric effects, respectively. The pyro-i effects

are almost the inverse effect: the creation of an induction ( $B$  or  $D$ , for pyromagnetic and pyroelectric materials, respectively) due to the variation in the temperature of the sample.

In addition, the difference between, for example, the pyroelectric effect and the thermoelectric effect must also be noted. Although both thermoelectric and pyroelectric effects are related to temperature, the thermoelectric effect is based on temperature gradients to produce a time-independent voltage (provided that the gradient remains time-independent), whereas the pyroelectric effect relies on the temporal evolution of temperature to produce the induction. In short, for thermo-i effects, a sample must be exposed to different temperatures at different points, whereas for pyro-i effects, a temperature change over time is required to produce the effect.

The first law of thermodynamics states that when an infinitesimal amount of heat  $dQ$  is exchanged with the material, and an infinitesimal amount of work  $dW$  is performed on or by the system, under quasi-static conditions, the corresponding infinitesimal change in internal energy  $dU$  is given as follows:

$$dU = dQ + dW \quad (1)$$

Assuming reversibility, we can write the following:

$$dU = TdS + \Phi_i dI_i \quad (2)$$

Using this set of variables, it is possible to calculate  $T$  and  $\Phi$  as follows:

$$T = \left( \frac{\partial U}{\partial S} \right)_I \text{ and } \Phi_i = \left( \frac{\partial U}{\partial I_i} \right)_S \quad (3)$$

which describe the thermal and ferroic equations of state, respectively.

In caloric and pyro studies, it is often convenient to define a different set of independent variables and instead use  $T$  and  $\Phi$ , as these are variables that can be readily controlled in the laboratory. The resulting thermodynamic function (or potential) is then the Gibbs energy, whose corresponding infinitesimal changes are as follows:

$$dG = -SdT - I_i d\Phi_i \quad (4)$$

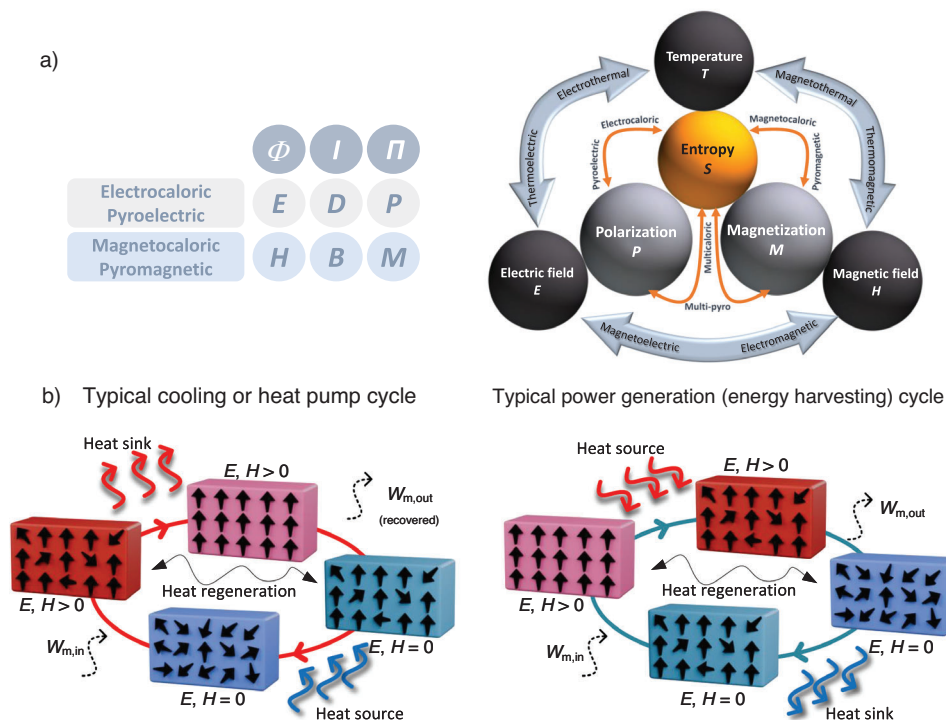
Using this set of independent variables, along with the Maxwell relation  $(\partial S/\partial \Phi_i)_T = (\partial I_i/\partial T)_{\Phi_i}$ , it is possible to calculate  $S$  and  $I$  as follows:

$$S = -\left( \frac{\partial G}{\partial T} \right)_{\Phi} \quad (5)$$

and

$$I_i = -\left( \frac{\partial G}{\partial \Phi_i} \right)_T \quad (6)$$

The above Maxwell relation represents the caloric/pyro coefficient of interest  $p_i = (\partial I_i/\partial T)_{\Phi_i}$ , which is often written as  $p_i = (\partial \Pi_i/\partial T)_{\Phi_i}$ , given that  $\Pi \approx I$ . To obtain large values of these coefficients, a significant temperature change of  $\Pi$  or  $I$  is required. The method of achieving this significant change can be a phase transition wherein each phase has significantly different  $\Pi$



**Figure 1.** a) Unified and specific thermodynamic notations for field, induction, and polarization for the considered i-caloric and pyro-i effects, along with an illustrative example of the relationship among magnetic field ( $H$ ), magnetization ( $M$ ), electric field ( $E$ ), polarization ( $P$ ), temperature ( $T$ ), and entropy ( $S$ ). The outer arrows denote the application domains related to coupling between fields and temperature. The inner arrows represent the physical interrelationships among external fields ( $\Phi = H, E$ ), polarization ( $\Pi = M, P$ ), temperature ( $T$ ), and entropy ( $S$ ). The pathways between them denote different, in this manuscript evaluated i-caloric and pyro-i effects. b) Illustration of the typical cooling or heat pump cycles and the typical power generation (energy harvesting) cycle.

or  $I$  values. These phase transitions can be of the first- or second-order type, and caloric effects and pyroelectric effects are typically greater near first-order phase transitions, where the polarization changes rapidly with temperature.

The equations presented above can be applied to various thermodynamic processes that constitute a thermodynamic cycle. For a thermodynamic cycle consisting of multiple processes, the integral of Equations (1) and (2) over the closed paths of these processes cancels out the internal energy, as it is a state function. Consequently, the first law of thermodynamics for a thermodynamic cycle can be expressed as follows:

$$\oint dQ = - \oint dW \quad (7)$$

or

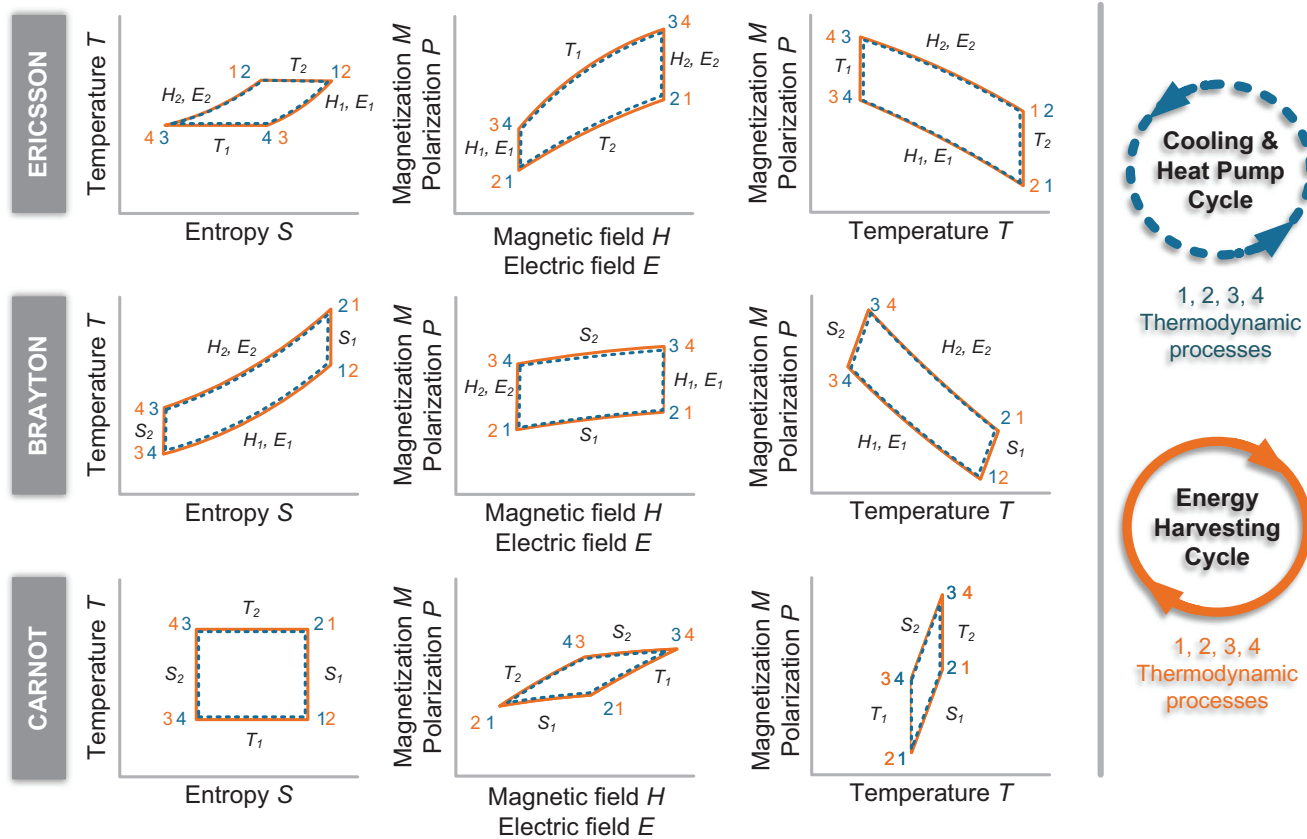
$$\oint TdS = - \oint \Phi_i dI_i \quad (8)$$

This indicates that the sum of all the heat absorbed or rejected from the observed sample equals the sum of all the work performed on or received by the sample. Figure 2 showcases various thermodynamic cycles frequently assessed in magnetocaloric and electrocaloric cooling and heat pumping, as well as in pyromagnetic or pyroelectric power generation. Note that there are numerous possible configurations of such cycles, including Stirling-type cycles (consisting of two isomagnetization or isopo-

larization processes and two isothermal processes), as well as combinations of different cycle types.

When conducting research beyond fundamental science, with the clear aim of applying new materials and functionalities to actual devices, it is necessary to establish metrics for comparing the performance of different designs and different materials. Therefore, it is crucial to understand key thermodynamic definitions that serve as figures of merit for caloric or pyro energy conversion technologies and related devices (Figure 3). Various definitions exist for the coefficient of performance ( $COP$ ) of caloric cooling ( $COP_C$ ) and heat pump ( $COP_H$ ) devices. To ensure a fair comparison between different technologies or devices, we should follow the conventional definition of  $COP$ , which is the ratio of the cooling or heating power to the total input power. This includes not only the magnetic, electric, or mechanical work for the caloric effect but also the total work input to all devices in the entire system. The  $COP$  is highly dependent on the temperatures of the heat sink and the heat source. Therefore, both terms must always be included in the definition when presented; otherwise, the presentation of the  $COP$  is insignificant. Excluding any form of work input, such as mechanical or hydraulic work, can mislead the community or the readers of a particular publication.

Note that the  $COP$  for heat pumps and cooling devices based on perfectly regenerative Ericsson and Stirling cycles can be defined in the same manner as that for the Carnot cycle. Similarly, the  $COP$  of the Carnot cycle operating under the same conditions as those of Ericsson or Stirling cycles can be evaluated. For cases



**Figure 2.** Various thermodynamic cycles in *temperature–entropy*, *polarization–field*, and *polarization–temperature* diagrams, with cycles for power generation depicted in orange and cycles for cooling or heat pumping shown in blue.

|                              | Energy                                 | Power                                        | Carnot cycle                          |
|------------------------------|----------------------------------------|----------------------------------------------|---------------------------------------|
| $COP_C$                      | $\frac{Q_C}{\sum W}$                   | $\frac{\dot{Q}_C}{\sum P}$                   | $\frac{T_C}{T_H - T_C}$               |
| $COP_H$                      | $\frac{Q_H}{\sum W}$                   | $\frac{\dot{Q}_H}{\sum P}$                   | $\frac{T_H}{T_H - T_C}$               |
| $\eta_{th}$                  | $\frac{W_{out}}{Q_{IN} + \sum W_{IN}}$ | $\frac{P_{out}}{\dot{Q}_{IN} + \sum P_{IN}}$ | $\frac{T_H - T_C}{T_H}$               |
|                              | Cooling                                | Heat pump                                    | Energy harvesting                     |
| $\eta_{ex}$                  | $\frac{COP_C}{COP_{C\_Carnot}}$        | $\frac{COP_H}{COP_{H\_Carnot}}$              | $\frac{\eta_{th}}{\eta_{th\_Carnot}}$ |
| Mass power density of device | $\frac{\dot{Q}_C}{m_{device}}$         | $\frac{\dot{Q}_H}{m_{device}}$               | $\frac{\eta_{th}}{\eta_{th\_Carnot}}$ |

**Figure 3.** Most key figures of merit for cooling, heating, and energy harvesting devices. The subscripts C, H, out, in, and ex represent cooling, heating, exit from the device, input to the device, and exergy, respectively; the subscripts Q, W,  $\dot{Q}$ , and P represent heat, work, heat rate (heating or cooling power), and power (mechanical or electrical), respectively; and COP and  $\eta$  represent the coefficient of performance and efficiency, respectively.



where the heat source and heat sink temperatures are related to gliding temperatures, such as in the Brayton-type cycle, the definition of the Carnot cycle, which is analogous to the operation parameters of the Brayton cycle, must be adjusted accordingly. This requires consideration of the entropy-averaged temperature of the heat supply and the heat absorption. Similar to the *COP*, the thermodynamic efficiency of energy harvesting cycles can be defined and compared with the thermodynamic efficiency of the analogous Carnot cycle.

An accurate definition of the thermodynamic efficiency of thermomagnetic and pyroelectric generators or motors considers the ratio of the generated power output, whether mechanical or electrical, to the heat input, including the power from all devices involved in the thermodynamic cycle. Any exclusions will lead to misleading information. An alternative definition of thermodynamic efficiency considers all energy inputs and electrical or mechanical work outputs, including all heat and any additional energy inputs required for the cycle. Moreover, the temperatures of the heat sink and heat source must always be included in the definition when presented; otherwise, the presentation of the thermodynamic efficiency is insignificant.

To compare the efficiencies of different technologies, regardless of their type of operation, the second law of thermodynamics introduces the principles of exergy and related exergy efficiency, which define how much valuable (transformable) energy is produced per unit of valuable (transformable) input energy. The exergy efficiency is occasionally also denoted as Carnot's efficiency or second law efficiency, and its simplest derivation for cooling and heating represents the ratio between the *COP* of the actual cycle and the *COP* of the Carnot cycle operating at the same temperatures of the heat sink and heat source. Similarly, the exergy efficiency of the power generation or energy harvesting cycle can be defined as the ratio between the actual thermodynamic efficiency and the thermodynamic efficiency of the Carnot cycle operating at the same temperatures of the heat sink and heat source.

One crucial figure of merit is the compactness of the devices or systems, known as power density. Notably, many researchers in the field of calorics often compare the power density of caloric devices (with the total power density of competitive technologies) based solely on solid refrigerants, neglecting the total mass of all devices and elements required to complete the desired thermodynamic cycle. This approach is flawed in terms of critical thinking and the message it conveys to future technological developers.

## 3. Materials

### 3.1. Magnetocaloric and Pyromagnetic Materials

The magnetocaloric characterization of materials has been extended in the literature, with a large variety of materials ranging from superconductors to molecular magnets and from oxides to metallic alloys. A comprehensive search in Web of Science for magnetocaloric-related publications from the last five years yields more than four thousand publications wherein "magnetocaloric," "magnetic entropy," "magnetic refrigerat\*," or "magnetic regenerat\*," are included as topics. However, not all the materials published in those papers will become part of a consumer or industrial device, or even a proof-of-concept prototype. There are two reasons for this decoupling between the engineer-

ing applications of thermomagnetic materials and the work of researchers. Magnetocaloric characterization has become a research tool for understanding the fine details of the phase transitions that materials undergo. This is crucial for unraveling the physics underlying the properties of the materials, with notable examples including the determination of critical exponents in multiphase materials where conventional techniques would not be effective,<sup>[29]</sup> or the rapid determination of phase diagram details for high-temperature skyrmion hosts,<sup>[30]</sup> to name a few cases. The other reason for the decoupling between engineering and magnetocaloric research is that many reported materials exhibit a relatively large isothermal entropy change, which is not accompanied by a large adiabatic temperature change when the material operates under cyclic conditions. The latter parameter is key to making a device viable. Materials with an adiabatic response below 2 K will not be efficiently usable in thermomagnetic devices.

In addition to having a sizeable magnetocaloric response, materials should exhibit additional secondary properties to be properly implemented in thermomagnetic devices, and a significant fraction of the abovementioned literature focuses on optimizing these properties. Among the most relevant factors, we should mention the following: corrosion resistance,<sup>[31–34]</sup> as the heat transfer fluid in actual devices might degrade the material; minimizing hysteresis (both magnetic and thermal),<sup>[35–42]</sup> because devices need to operate in cycles of field and temperature; fast time response and optimized dynamics of the phase transformation,<sup>[43,44]</sup> since delays in the response of the material to excitation would decrease the operational frequency of the device; formability or machinability, to configure the required parts of the thermomagnetic device in such a manner that it can be efficiently integrated with the rest of the device; and mechanical stability, such that the active material is not affected by the repetitive phase transition that might lead to its eventual disintegration.<sup>[45–54]</sup>

In recent years, extensive review papers have been published on the magnetocaloric responses of materials,<sup>[55–65]</sup> either with broad coverage or a focus on specific families. Rather than repeating the extensive tables of data and references, we focus here on the main aspects of the research on the most active magnetocaloric materials. In the last five years, the most investigated families of magnetocaloric materials, according to the number of publications appearing in scientific literature, are presented in **Figure 4a**. There is a predominance of manganites and other complex oxides,<sup>[56]</sup> of which rare earth-transition metal oxides could be suitable for low-temperature applications. However, the comparison of the magnetocaloric response per volume of the sample could be less favorable than for other alloys and compounds. Most studies use magnetocaloric characterization to understand the nature of their phase transition without actual perspectives to implement these materials in actual devices.

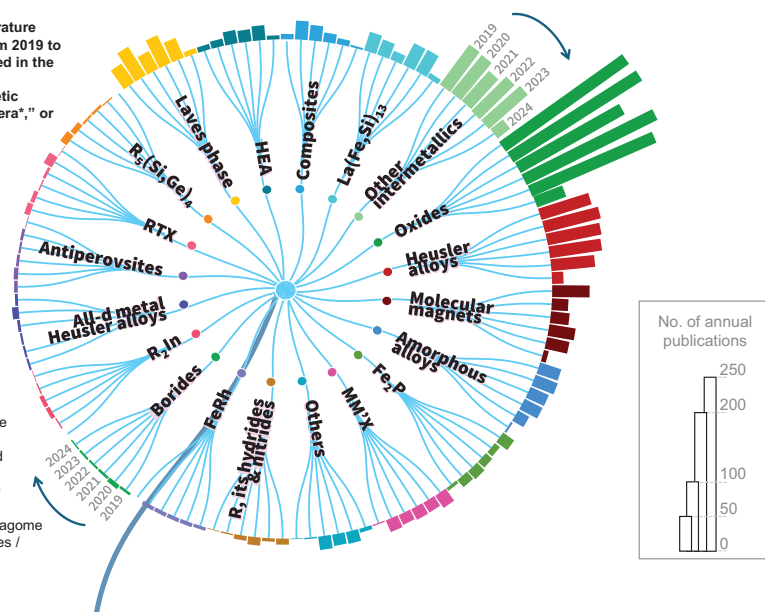
Second in importance are Heusler and half-Heusler alloys, with  $X_2YZ$  and  $XYZ$  stoichiometries, respectively. The X and Y sites are transition metals, whereas Z is an element from groups 13 to 15 of the periodic table. From a magnetocaloric perspective, a typical example is  $Ni_2MnZ$ , with  $Z = Ga, In, Sn, \text{ or } Sb$ , which undergoes a magnetostructural transition from cubic ferromagnetic austenite at high temperatures to tetragonal paramagnetic or ferrimagnetic martensite at temperatures below the

### a) Fraction of annual publications

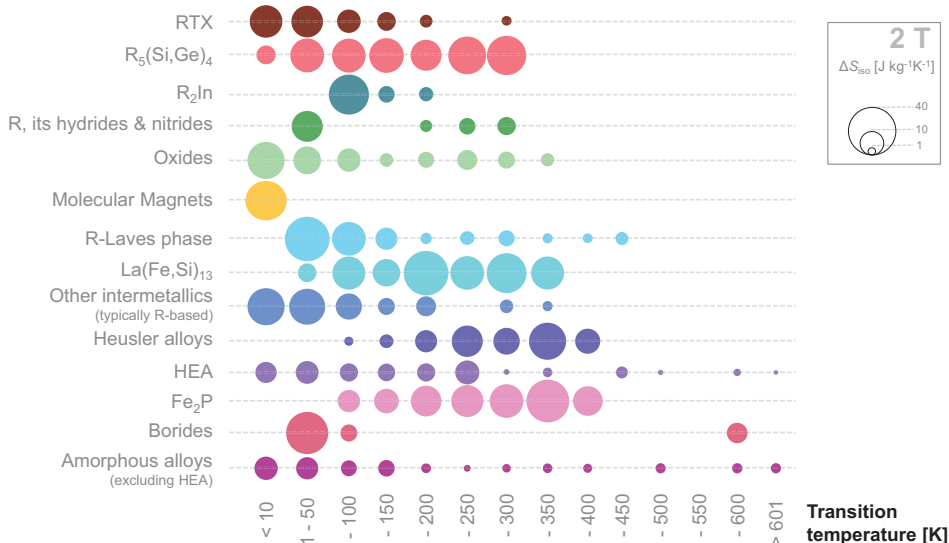
Web of Science (WoS) literature survey was performed from 2019 to March 2024. The terms used in the search fields were: "magnetocaloric," "magnetic entropy," "magnetic refrigerator," or "magnetic regenerator."

R represents rare-earth elements, T as transition metallic elements, X as metalloids, and HEA as High-Entropy Alloys. MM'X family: M and M' represent Mn, Ni, Fe, or Co. Oxides consist of ferrites, spinels, double perovskites, manganites, garnets, borates, etc.

Other intermetallics comprise  $Mn_2X$ ,  $MnX$ , clathrates, and many other rare-earth based intermetallics not separately shown. Others represent 2D materials, Weyl semimetals, organic-inorganic hybrids, Kagome lattices, or 3d or 4f complexes / clusters.



### b) Comparison matrix for $\Delta S_{iso}$ magnetocaloric materials of rising popularity



**Figure 4.** a) Overview of the various families of magnetocaloric materials reported during recent years. A Web of Science (WoS) literature survey was performed up to March 2024. R represents rare-earth elements, T indicates transition metallic elements, X denotes metalloids, and HEA indicates high-entropy alloys. MM'X family: M and M' represent Mn, Ni, Fe, or Co. Oxides consist of ferrites, spinels, double perovskites, manganites, garnets, borates, etc. Other intermetallics comprise  $Mn_2X$ ,  $MnX$ , clathrates, and many other rare earth-based intermetallics not separately shown. Others represent 2D materials, Weyl semimetals, organic-inorganic hybrids, Kagome lattices, or 3d or 4f complexes/clusters. b) Isothermal entropy change ( $\Delta S_{iso}$ ) versus the transition temperature of selected families, based on data from.<sup>[58,60,66,67]</sup> The magnitude of  $\Delta S_{iso}$  (for 2 T) is represented by the size of the circles.

transition temperature.<sup>[68,69]</sup> This structural change, which is associated with a change in the magnetic character, produces a significant magnetocaloric response. However, not all Heusler alloys undergo a martensitic transition. There are those with exceedingly high Curie temperatures that do not coincide with their structural transitions, limiting their applicability as magnetic refrigerants. The tunability of the composition, either by tuning

the stoichiometry or by substitution, leads to broad tunability of the transition temperatures, ranging from moderately low to moderately high temperatures. Despite their advantages, one of the main challenges of Heusler alloys is decreasing their large thermal hysteresis, which is typically accomplished by increasing the geometrical compatibility between the martensite and austenite phases. These alloys have served as testing grounds

for the development of Temperature First Order Reversal Curves techniques,<sup>[70,71]</sup> which are useful for predicting the operating conditions of magnetocaloric materials under partial transformation when incorporated into actual engineering devices. The significant volume change that occurs when the material undergoes a first-order phase transition usually causes cracks in the samples, leading to a loss of mechanical integrity after cyclic operation. This effect is also deleterious to engineering applications.

All-d-metal Heusler alloys merit separate mention because they are considered an alternative for enhancing the mechanical integrity of Heusler alloys. They are derived from the Ni(Co)–Mn–Ti composition<sup>[72]</sup> and exhibit a large magnetic entropy change near room temperature after proper compositional modifications, which generally include the partial substitution of Ni by Fe or Co. Despite their large magnetic entropy change values and good mechanical properties, their primary drawback is the low cyclic response caused by their large thermal hysteresis (on the order of 10 K), which limits their practical applicability.

A significant amount of research has been conducted on La(Fe,Si)<sub>13</sub>, which is primarily based on elements that are not critical.<sup>[73]</sup> This material has remained prominent due to the tunability of its transition temperature, achieved not only by compositional changes but also via hydrogenation.<sup>[74]</sup> The first-order phase transition that this material undergoes is magnetoelastic; that is, it is not accompanied by a crystallographic symmetry change but by a change in lattice volume, which poses advantages over magnetostuctural transformations from a mechanical perspective. Compositional changes such as the modification of the Fe:Si ratio lead to a change from a first-order phase transition to a second-order phase transition.<sup>[75]</sup> At the critical point of the phase transition, also known as the tricritical point in this context, the material should retain the best characteristics for both types of transitions, such as a large magnetocaloric response without hysteresis. Instead of a trial-and-error process to determine this optimized composition, predictive techniques can be used.<sup>[76]</sup>

MM'X alloys, with M and M' being transition metals and X a main group element, share some of the properties of half-Heusler alloys. Specifically, they undergo a first-order magnetostuctural phase transition from low-temperature orthorhombic to high-temperature hexagonal.<sup>[77–79]</sup> The primary challenge in identifying magnetocaloric compounds in this family is tuning the transition temperature to be in the range between the Curie temperatures of the martensitic and austenitic phases. This produces a significant change in magnetization upon phase transformation, leading to a large magnetocaloric effect. Most materials in this family respond near room temperature, although they have drawbacks similar to those of Heusler alloys (hysteresis and limited structural integrity upon cycling).

Compounds derived from Fe<sub>2</sub>P, such as (Mn,Fe)<sub>2</sub>(P,Si), have attracted significant attention in recent years.<sup>[80]</sup> They also undergo a magnetoelastic transition from a low-temperature ferromagnetic phase to a high-temperature paramagnetic phase. The absence of critical elements and avoidance of poisonous elements, such as As, which was present in the original compositions, favored this line of research. Compositional variations also enable tuning of the transition temperature in an environment at room temperature and alter the nature of the phase transition to the tricritical point, achieving very low thermal hysteresis values.<sup>[81]</sup>

The search for optimized mechanical properties has led researchers to explore the HEA space.<sup>[59]</sup> Initially, publications in this area exhibited a significantly low magnetocaloric response because they were restricted to the most traditional definition of a high entropy alloy, focusing on equiatomic compositions.<sup>[82]</sup> Thus, the dilution of the magnetic moments of the constituent elements produced a reduction in the isothermal entropy change with respect to medium- or low-entropy alloys. The scenario significantly changed when, by using a different search strategy that starts from medium-entropy alloys with a first-order phase transition, the high-entropy alloy region is reached by isostructural substitutions.<sup>[83]</sup> Thus, it is possible to bridge the gap between the magnetocaloric responses of HEAs and those of high-performance magnetocaloric materials.<sup>[84]</sup> The primary advantage of HEAs, in addition to their potentially good mechanical properties, is that there is an extremely large search space that allows the optimization of different complementary properties. Similarly, amorphous alloys provide significant flexibility in compositional tuning, rendering it easy to adjust the transition temperature to the desired range. However, crystalline HEAs exhibit significant stability against temperature-induced irreversible transformations, which could potentially make them superior to amorphous alloys. The combination of a large search space, the hope of enhanced mechanical properties, and the demonstration of significant values of the magnetocaloric response when the proper search strategy is followed makes this a field that will likely continue to grow in the coming years.

Laves phases are intermetallics with AB<sub>2</sub> stoichiometry that crystallize in cubic MgCu<sub>2</sub>-hexagonal MgZn<sub>2</sub>- or cubic MgNi<sub>2</sub>-type crystal structures. From a magnetocaloric perspective, the most relevant series are RNi<sub>2</sub>, RCo<sub>2</sub>, and RAl<sub>2</sub>.<sup>[85,86]</sup> The order of their phase transitions has been controversial in certain cases, especially for DyCo<sub>2</sub>, motivated by very low hysteresis and the misleading application of the Banerjee criterion, which helped resolve this controversy.<sup>[87]</sup> The main interest in this family lies in the cryogenic range, with several compositions having a significant response in the liquid hydrogen temperature and a large tunability of the transition temperature by combining two different rare earths.<sup>[88]</sup> In the case of RAl<sub>2</sub>, in addition to the thermomagnetic phase transition, a spin reorientation transition occurs at lower temperatures that broadens the magnetocaloric response without decreasing the magnitude of the primary peak. This could be considered a suitable feature for enhancing the performance of refrigerators at lower temperatures. In addition to these well-known families of Laves phases, there are others that have been significantly less studied, such as RMn<sub>2</sub><sup>[89]</sup> and RMg<sub>2</sub>.

Another emerging research field involves the use of molecular magnets in magnetocaloric refrigeration.<sup>[63,64,90]</sup> This builds upon pioneering work in developing nanomagnets for magnetic refrigeration,<sup>[91]</sup> integrating the advantages of well-defined size, shape, and magnetic moment of molecular magnets, which surpass those of inorganic superparamagnetic nanoparticles composed of rare-earth and transition-metal elements. Their molecular characteristics, including molecular anisotropy and molecular spin degeneracy, facilitate the tuning of their magnetic properties in the ultra-low temperature range.

Regarding the temperature range, most of the literature focuses either on room-temperature magnetocaloric materials for refrigeration and air conditioning or on the temperature range

suitable for gas liquefaction due to the increasing importance of hydrogen as an alternative energy vector. This is observed when we present an overview of the magnetocaloric effects of various material families versus their transition temperatures, as shown in the right panel of Figure 4. We used isothermal entropy change ( $\Delta S_{\text{iso}}$ ) to represent the magnetocaloric effect of the materials, as it is the most widely characterized parameter reported in the literature and may help complete this survey of the operation temperatures for magnetocaloric materials. When addressing higher-temperature materials, most well-known magnetocaloric materials decrease in magnetocaloric performance when the transition temperature is increased. A significant exception is the tuning of the magnetocaloric response of transition-metal-based amorphous alloys.<sup>[92,93]</sup> With a high content of magnetic elements, tuning the transition temperature to wider operation regimes is achieved by further increasing this content, which leads to a larger change in magnetization. If this is performed without changing the critical exponents of the second-order phase transition, the outcome is a larger isothermal entropy change when the transition is shifted to higher temperatures. Unfortunately, owing to their second-order characteristics, these amorphous alloys might be at the borderline of usability for temperature control devices. However, they may also be valuable for applications in thermomagnetic energy conversion.

Metal monoborides such as MnB have also been studied for high-temperature applications. Replacing Mn with other metals, such as Fe and Co, allows for tuning the transition temperature over a broad range, although it significantly reduces the magnetocaloric response.<sup>[94]</sup> However, if substitution occurs at the metalloid site, such as in  $\text{MnB}_{(1-x)}\text{Ge}_x$  and  $\text{MnB}_{(1-x)}\text{Si}_x$ ,<sup>[95]</sup> which are initially screened using computational techniques, it is possible to increase the magnetocaloric response while maintaining their transition temperatures at  $\approx 600$  K. However, challenges remain in the production of these samples as single-phase materials.

### 3.2. Electrocaloric and Pyroelectric Materials

Pyroelectric materials, which generate a change in electric induction in response to a change in temperature, and electrocaloric materials, which generate a change in temperature in response to a change in electric induction, have been extensively studied in the last 50 years. This interest is partly due to their applications in infrared imaging and sensing, as well as their more recent potential for energy harvesting from waste heat and cooling.<sup>[96–101]</sup> During this period, two main categories of materials have stood out: ferroelectric oxides and ferroelectric polymers. In this review, we focus on these two categories and do not review ferroelectric liquid crystals<sup>[102]</sup> or molecular ferroelectrics.<sup>[103]</sup>

Traditional pyroelectric and electrocaloric materials are largely lead-based. However, owing to environmental concerns and regulatory limitations, there has been a shift toward lead-free alternatives. These new materials include various ceramics and polymers that have been engineered to exhibit similar or enhanced pyroelectric and electrocaloric properties, without the toxicity issues associated with lead. Lead zirconate titanate is one of the most widely utilized pyroelectric oxides. It exhibits high pyroelectric coefficients at elevated temperatures and has been incorporated into various composites and forms to optimize its prop-

erties for different applications. This includes the formation of composites with ferroelectric polymers, such as polyvinylidene fluoride, to enhance pyroelectric properties and fabricate flexible sensors. Their pyroelectric properties have also been improved via processing methods, such as hot pressing, which affects the microstructure and electrical behavior of the composites.<sup>[104]</sup>

Barium titanate is widely used in pyroelectric applications due to its large pyroelectric coefficient near room temperature and is particularly suitable for pyroelectric photodetectors owing to its narrow bandgap. Barium titanate is often used in its pure form or in combination with other materials to enhance its properties. For instance, barium titanate nanoparticles have been incorporated into polyvinylidene fluoride polymeric matrices to improve the pyroelectric response of composites, primarily by increasing the crystallinity of the ferroelectric  $\beta$ -phase of polyvinylidene fluoride, which is crucial for its electroactive properties. Barium titanate displays a robust electrocaloric response near and above room temperature owing to its first-order electrostructural phase transitions.<sup>[105,106]</sup> This oxide exhibits high electrocaloric strength in single-crystalline form.<sup>[107]</sup> Other oxides that have been investigated for pyroelectric and electrocaloric applications include solid solutions of lead magnesium niobate and lead titanate near their morphotropic phase boundaries,<sup>[108]</sup> as well as lead scandium tantalate,<sup>[109]</sup> sodium bismuth titanate,<sup>[110]</sup> and strontium bismuth tantalate.<sup>[111]</sup>

Polyvinylidene fluoride can crystallize into several phases, with the ferroelectric  $\beta$ -phase being central for its pyroelectric and electrocaloric properties. Techniques such as mechanical stretching, electrical poling, co-polymerization, and the inclusion of nanoparticles are used to enhance these properties. Research into these fluorinated polymers focuses not only on enhancing their intrinsic properties but also on developing composites that combine the benefits of oxides and polymers.

Recent studies have focused on enhancing the robustness, reversibility, and energy efficiency of pyroelectric and electrocaloric materials via advanced manufacturing techniques. These techniques include epitaxial thin-film growth, which allows for precise control over material characteristics at the microscopic level,<sup>[112,113]</sup> strain engineering,<sup>[114]</sup> and multilayer fabrication,<sup>[115]</sup> which enables the application of large electric fields and, consequently, large pyroelectric and electrocaloric effects. For example, multilayer capacitors composed of ferroelectric lead scandium tantalate display electrically driven changes (5.5 K) near room temperature, and the temperature change remains large over a wide range of temperatures, which is necessary for various applications.<sup>[116]</sup> The key step is to drive the first-order ferroelectric phase transition in these multilayer capacitors using supercritical electric fields without breakdown. Multilayer capacitors have now been used in several electrocaloric and pyroelectric devices.<sup>[117–119]</sup>

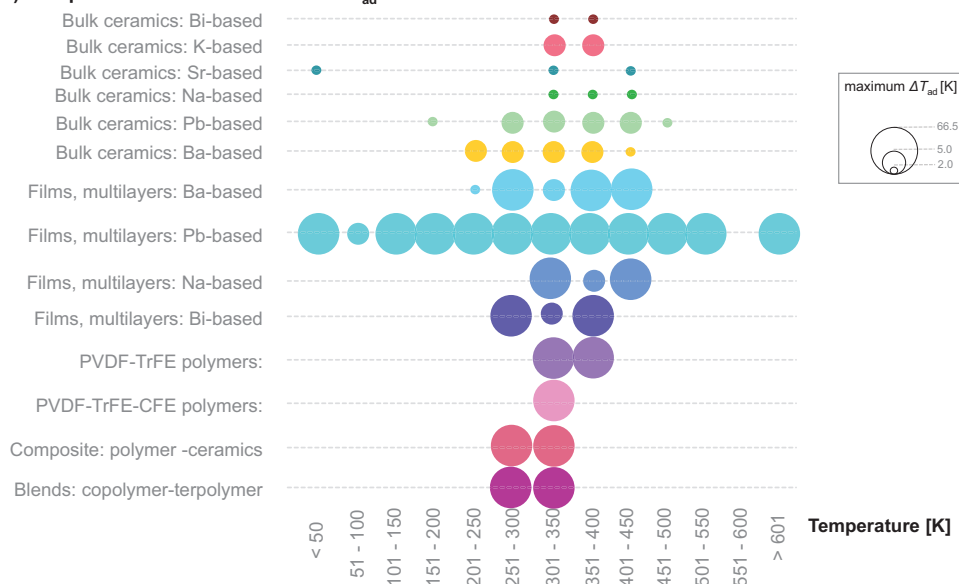
In addition to the experimental data shown in Figure 5, certain theoretical studies on high-temperature electrocaloric materials have been reported. For example, Zhang et al.<sup>[99]</sup> theoretically modeled the electrocaloric effect in ferroelectric and antiferroelectric materials, demonstrating high-temperature and significant electrocaloric effects for the bulk and nanowires of  $\text{PbTiO}_3$ ,  $\text{BaTiO}_3$ , and  $\text{KNO}_3$ , with peak adiabatic temperature changes at  $\approx 600$ , 500, and 750 K, respectively. These changes were more than one order of magnitude higher than those reported for the



**a) Comparison matrix for  $\Delta T/\Delta E$  of electrocaloric materials**



**b) Comparison matrix for maximum  $\Delta T_{ad}$  of electrocaloric materials**



**Figure 5.** a) Overview and comparison of the ratio of adiabatic temperature change to electric field change for various families of electrocaloric materials reported during recent years, based on data from refs.[121–130] The magnitude of  $\Delta T/\Delta E$  is presented by the size of the circles. b) Comparison matrix of the maximum adiabatic temperature change for the various families of electrocaloric materials, as presented in a), and reported during recent years, based on data from refs.[121–130] The magnitude of  $\Delta T_{ad}$  is indicated by the size of the circles.

same materials near room temperature. A similar study<sup>[120]</sup> performed using molecular dynamics and first principles showed a giant electrocaloric effect of lithium niobate,  $\text{LiNbO}_3$ , peaking at  $\approx 1600$  K. Moreover, in this case, the reported adiabatic temperature change was at least one order of magnitude higher than that at room temperature.

Although these two studies, among several other similar studies, represent a theoretical evaluation of the electrocaloric effect at high temperatures, they convey three crucial messages related to future research activities:

- The electrocaloric effects of certain materials can be an order of magnitude higher than those at room temperature.
- Material scientists should focus on experimentally evaluating the electrocaloric effect, at least for reported theoretical works.
- If experimentally proven, high-temperature electrocaloric materials can bring about a realistic breakthrough in the domains of ultrahigh-temperature heat pumps and small appliances related to high-temperature pyroelectric energy harvesting.

The selected and experimentally investigated families of electrocaloric materials that appear in the scientific literature are presented in Figure 5.

### 3.3. Multicaloric and Multipyro Materials

Multicaloric materials undergo thermal changes that can be driven simultaneously or sequentially by more than one type of external field.<sup>[131]</sup> In this review, we focus on multicaloric materials that can be driven using magnetic and electric fields (i.e., multiferroic and magnetoelectric materials).<sup>[132]</sup> The combination of these two driving fields can be beneficial for device design from different perspectives. One evident use would be the multimode operation of the material, as the phase transition could be driven by one of the two driving fields depending on the operating conditions (e.g., using the electrocaloric effect in one temperature range and the magnetocaloric effect in another). This would lead to larger thermal changes. Alternatively, if the composite material exhibits both transitions in the same temperature range, the use of these two fields can reduce the operating fields. In addition, when phase transitions are of the first-order type, it would be possible to ameliorate hysteresis losses by a suitable sequence of application of the two fields, transferring the hysteresis from one polarization to the other.

Single-phase materials that exhibit large magnetocaloric+electrocaloric or pyromagnetic+pyroelectric responses simultaneously are rare. Therefore, the predominant approach has been to develop composites from phases that individually exhibit one of the desired responses. Over the last decade, several composite materials that are both magnetocaloric and electrocaloric have been developed. These include LCMO/BTO,<sup>[133]</sup> Fe–Rh/BaTiO<sub>3</sub>,<sup>[134]</sup> and Fe–Rh/PMN–PT.<sup>[135]</sup>

A material that can develop changes in magnetization and ferroelectric polarization in response to a change in temperature would be characterized as a multi-pyro material. Pyromagnetic and pyroelectric coefficients have not been reported for the same material because multiferroic materials that exhibit both ferromagnetism and ferroelectricity are rare. However, the composites of magnetocaloric and electrocaloric materials described above should exhibit multi-pyro effects. While there is a significant amount of literature on materials exhibiting individual caloric or pyro effects, the activities of the community have been less intense for multicaloric and multi-pyro materials that combine electric and magnetic fields. This is motivated not only by the more complicated preparation of composites with suitable coupling between the transitions but also by the more complex characterization techniques required to comprehensively evaluate these materials. As potential device designs, operation cycles, and prototypes will lead to more significant improvements in performance owing to the multicaloric/multi-pyro nature of the materials, we expect increased research efforts in this area.

### 3.4. A Note on the Comparison of Different Caloric Materials

Researchers studying caloric materials usually try to make two types of comparisons: intracaloric effects and comparisons between different caloric technologies. The first type may seem sim-

ple: it involves collecting all published information about a particular caloric effect (e.g., magnetocaloric, electrocaloric, etc.) and presenting the results using an Ashby plot. However, despite the conceptual simplicity, several challenges need to be addressed. One major hurdle is that, depending on the characterization technique used (e.g., direct, quasi-direct, indirect methods or extrapolations), publications tend to report either the isothermal entropy change or the adiabatic temperature change and only a few publications include both measurements. Although this situation has improved recently, it is not yet fully resolved. Moreover, with advances in computer power and methods, several studies are now purely theoretical and rely on approaches such as molecular dynamics and first-principles calculations. The results from such studies are to a lesser extent verified by the experiments.

Another important challenge is that authors often report the caloric response at the maximum achievable excitation, which can lead to unrealistic performance evaluations of caloric materials in practical applications. In electrocaloric studies, measurements are usually performed up to the maximum excitation until the voltage breakdown of the material. Similarly, in mechanocalorics, it is highly impractical to apply excessively large forces approaching the limits of material breakdown. As mentioned in this manuscript, a practical device must be capable of performing at least 10<sup>8</sup> thermodynamic cycles during its lifetime. Furthermore, for certain technologies, such as mechanocalorics, it is difficult to achieve high frequencies – especially those higher than 5 Hz (required for the compactness of the device) – while maintaining very high load values. This difficulty arises from the fact that the machines required to generate such significant voltage changes are generally very robust and not capable of delivering very short time intervals.

Importantly, unlike some magnetocaloric studies, and to the best of our knowledge, none of the studies on electrocaloric or mechanocaloric materials have demonstrated the dependence of the caloric effect on cycle frequency. It is known that for some materials, cycling (compared to single-shot experiments) can significantly affect the caloric effect. Unfortunately, information on this topic is mainly available in the context of magnetocalorics, with a notable example presented by T. Gottschall et al.<sup>[5]</sup> High operating frequencies are essential for all caloric materials in order to develop devices with high power density relative to the total mass of the device, which is crucial for market competitiveness. Another important consideration is the impact of operating frequency on the hysteresis of a material. To the best of our knowledge, this important relationship has not yet been adequately studied in the context of caloric materials. A proper evaluation of the theoretical performance of caloric materials requires addressing the issues mentioned above. In addition, this information must be included in the theoretical evaluation of the thermodynamic cycle. Furthermore, the only valid method to evaluate the thermodynamic cycle of a caloric material is the availability of reliable data on the temperature-entropy field dependence. Unfortunately, this information is not readily available for most caloric materials.

In some manuscripts, the authors compare material properties based on COP. The COP is defined as the ratio of cooling or heating power to total working power and serves as a measure of the performance of an appliance. In conjunction with precisely defined temperatures for the heat source and the heat sink, the

COP provides valuable information that can be compared with the analogous Carnot cycle. However, the theoretical evaluation of the COP for material is highly problematic if it does not take into account the realistic temperature-entropy field dependency and the challenges mentioned above. Furthermore, such assessments often ignore the characteristics of the potential field source and its efficiency, as well as the performance of heat recovery and work recovery. Caloric technologies differ significantly in these aspects, and a comparison that does not take these differences into account can mislead the reader. Another problem with the definition of COP for materials is the idealization of thermodynamic cycles. Experts in the field know that it is possible to generate different thermodynamic cycles, not only theoretically, but also through the specially tuned operation (field and heat transfer) of a caloric device. If the comparison of materials is based solely on their theoretical and idealized properties, it is easy to show that all theoretically evaluated materials can perform the ideal Carnot cycle. Therefore, comparing ideal thermodynamic cycles between different materials is meaningless since the ideal Carnot cycle is fully reversible and has an exergy efficiency of 100. In other words, comparing the ideal Carnot COP of a magnetocaloric device with the ideal Carnot COP of an electrocaloric device is not meaningful, as both are ideal and represent the best possible performance. Furthermore, when evaluating thermodynamics, one should distinguish between different configurations such as single-stage, cascaded, and regenerative processes as well as their possible configurations involving thermal diodes or thermal switches. Not all materials and the principles of their use fall into exactly the same category.

From a practical point of view, manufacturability and thermal properties are an important aspect when comparing different materials. While magnetocaloric materials exhibit moderate thermal diffusion and mechanocaloric materials largely overcome this challenge, electrocaloric materials, especially polymers, have very poor thermal properties. However, if the electrodes are suitably designed and integrated into multilayer structures of electrocaloric materials, this can significantly increase the effective thermal diffusion capability, but also impair the overall electrocaloric effect. In addition, the way the materials are processed affects the heat transfer area of the regenerators, which should be maximized. The processing methods and stages of development vary between different caloric materials and technologies and have a significant impact on the overall performance of a device. Therefore, comparing materials without considering these aspects is also highly problematic.

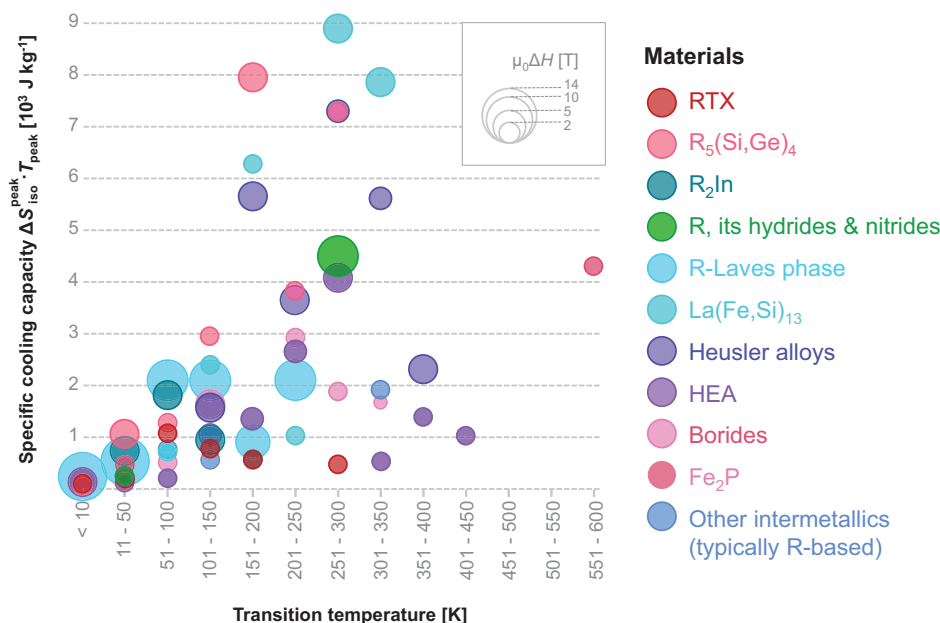
In conclusion, we advise the caloric community to avoid comparisons that ignore the above aspects. Such comparisons can mislead readers, create an unfair bias toward certain technologies, and do not accurately reflect the performance characteristics that are particularly important for newcomers to the field to understand the materials and corresponding devices. We encourage researchers to base their comparative analysis on the following criteria:

- For a realistic assessment of materials that serve as a basis for practical applications, information on the temperature-entropy field is essential. This data must take into account the dynamic caloric effect and the hysteresis associated with the operating frequency.
- Material properties should only be specified for field strengths that allow the material or its structure to operate beyond  $10^8$  thermodynamic cycles and for realistically applicable field sources for the intended application. If these conditions are not met, any comparison is highly questionable.
- Since different caloric materials have different thermal properties and can only be tailored by specific processing methods, the only fair comparison is to use processing-related characteristics of these materials.
- A realistic comparison between different materials requires that the properties are compared on the basis of the same characterization method. If this is not possible, it must always be stated which method was used to characterize a particular material. In addition, the properties of caloric materials determined solely by theoretical evaluation should not be compared with those derived from experimental results.
- Researchers should endeavor to provide comprehensive carbon footprint information for each material. When comparing the carbon footprint with other solid or liquid refrigerants, it is important to ensure that the comparison is fair and unbiased. This means avoiding the promotion of one technology at the expense of another by comparing the best refrigerant of one technology with inferior or even banned refrigerants of another.
- Researchers should avoid calculating and comparing only the coefficient of performance (COP) of a material, as such an analysis does not provide enough information for meaningful comparisons. If this type of comparison is nevertheless carried out, researchers should adhere to the criteria mentioned above and take into account the problematic aspects discussed in this subchapter.

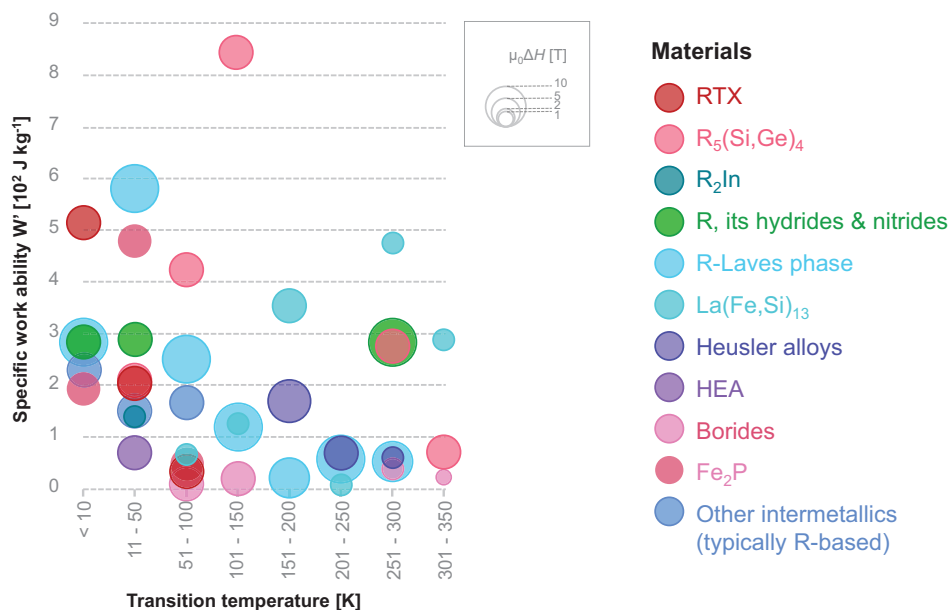
Taking into account all of the above and the information available on materials — some of which may be incomplete — the representation of material properties in the same diagrams should contain simple definitions. These can be derived from isothermal entropy and adiabatic temperature changes. This approach can be extended to include a simple definition of the relative cooling or heating capacity, calculated as the product of the temperature and the isothermal entropy change for the applied field, and a simple definition of the working absorption potential, which is derived from the product of the adiabatic temperature change and the isothermal entropy change for the applied field. Even if this information does not provide a solid basis for comparing caloric materials due to the problems already mentioned, it can at least show which material groups perform well in relation to the application temperature. In addition, it can reveal gaps where certain material groups are lacking in certain temperature ranges or highlight areas that require further research in each area.

Figure 6 shows the specific cooling capacity (Figure 6a) and specific work absorption ability (Figure 6b) of different types of magnetocaloric materials. Since unfortunately, no information is available on adiabatic temperature changes, the temperature dependence for the specific cooling capacity is shown more comprehensively. Most of the data are given for temperatures up to about 350–400 K. This does not necessarily mean that there are no efficient magnetocaloric materials at higher temperatures, but rather reflects the current focus of research activities in materials science. The specific cooling capacity of low-temperature

a) Specific cooling capacity for magnetocaloric materials



b) Specific work ability of magnetocaloric materials



**Figure 6.** The specific cooling capacity (6a) and the specific work absorption ability (6b) as a function of the temperature and the applied field for different magnetocaloric materials (refs.[58,81,136,137,65]).

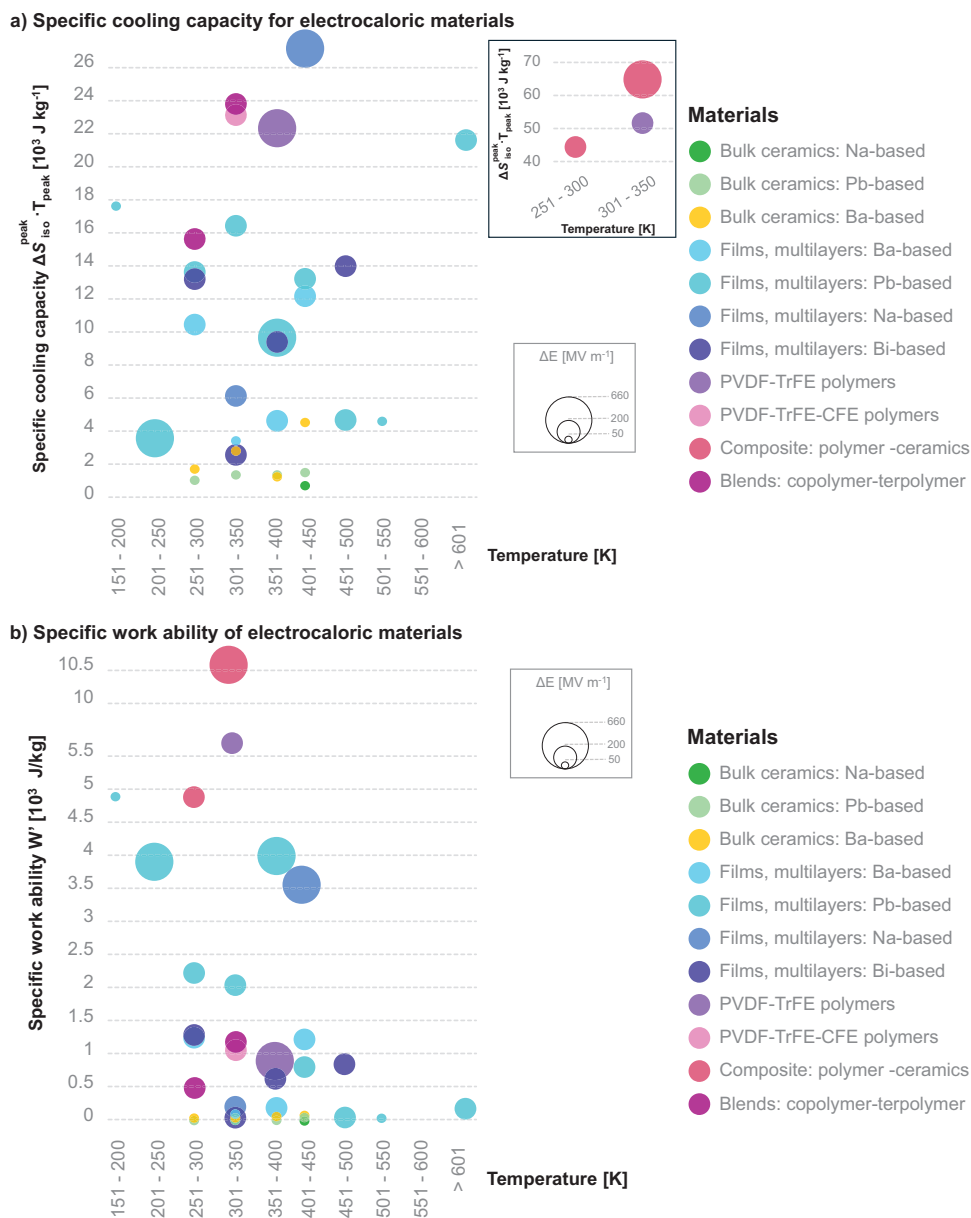
refrigerants is limited due to the low temperature itself. However, the specific work absorption ability of these materials is generally higher than that of materials operating at higher temperatures.

**Figure 7** illustrates the specific cooling capacity (Figure 7a) and the specific work absorption ability (Figure 7b) of different types of electrocaloric materials.

Unfortunately, due to the lack of available information on isothermal entropy changes, the temperature dependence of

the specific work absorption ability is shown more comprehensively. Most of the data are given for temperatures up to about 450–500 K. This does not necessarily mean that there are no high-performance electrocaloric materials at higher temperatures, but rather reflects the current focus of research activities in materials science. On the other hand, the existing literature for cryogenic temperatures does not contain information on electrocaloric materials with performance comparable to those shown in Figure 7. The specific work absorption ability of these





**Figure 7.** The specific cooling capacity 7a) and the specific work absorption ability 7b) as a function of the temperature and the applied field for different electrocaloric materials (refs.[122,127–130]).

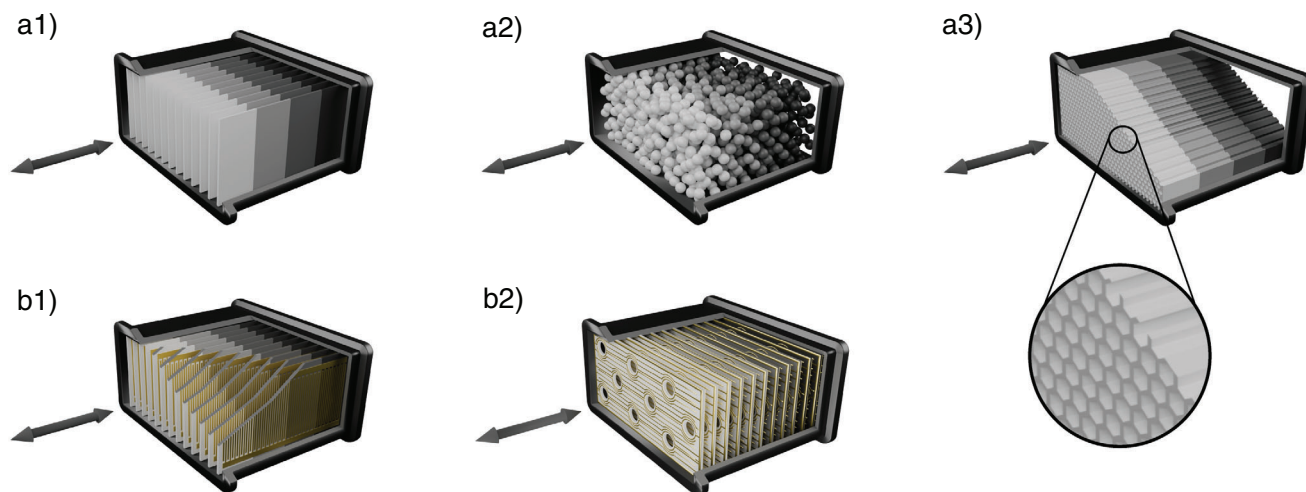
materials is highest for materials in the range between 200 and 450 K.

## 4. Devices and Applications

The latest and most comprehensive reviews on different types of devices, their design, and operation characteristics can be found in the following references and related domains: magnetocalorics,<sup>[138]</sup> pyromagnetics,<sup>[138,28]</sup> electrocalorics,<sup>[98,139]</sup> and pyroelectrics.<sup>[140–143]</sup> Unfortunately, for domains of multicalorics and multi-pyro, no prototype device related to the combination of the aforementioned phenomena and/or domains has been developed, except for a few experimental

attempts or simulations.<sup>[144]</sup> In this review, we address the high-quality published literature on caloric materials and devices (magnetocalorics,<sup>[23,24]</sup> electrocalorics, pyroelectrics,<sup>[25–27]</sup> and barocalorics).<sup>[145]</sup>

Further recommended research that addresses all caloric devices and related problems can be found in Schipper et al.<sup>[146]</sup> who performed a comparative study on the efficiency of 36 different caloric materials and compared them to the exergetic grades of compressors. A review of numerical modeling approaches and optimization possibilities for all types of caloric technologies was provided by Silva et al.<sup>[147]</sup> Another study evaluated different caloric materials and their performance in residential heat pumps.<sup>[148]</sup> Torello and Defay<sup>[149]</sup> conducted an insightful



**Figure 8.** Active caloric regenerators: a1–a3) magnetocaloric; b1–b2) electrocaloric. Arrows denote the oscillation of the fluid flow. Different colors of regenerators along the fluid flow path indicate tuned caloric materials according to their Curie temperature. a1) Ordered structure of layers of parallel magnetocaloric plates. a2) Packed bed of grains of different magnetocaloric materials, layered similarly to the ordered structure. a3) Ordered honeycomb or similar structure of layered magnetocaloric materials. b1) Parallel multilayered, multielectrode structures of electrocaloric plates. b2) Potential design of the multilayered, multielectrode structure of electrocaloric materials with voids for the oscillating fluid flow.

statistical study on the performance of caloric prototypes published to date. Accordingly, they defined an effective heat transfer coefficient for parallel-plate and spherical particle-based regenerators with different porosities and caloric materials. Masche et al.<sup>[40]</sup> studied hysteresis effects on six fictive caloric materials with different properties and found a stronger negative impact on the COP than on the cooling power for a potential regenerator. They noted that materials with low isothermal entropy changes are more affected by hysteresis, whereas those with low specific heat can reduce their impact on regenerator performance. A comprehensive comparative study on the performance of different heating and cooling caloric systems was conducted by Johra,<sup>[150]</sup> who compared the temperature- or power-dependent performances of caloric devices and conventional vapor compression devices. They also provided an interesting overview of the historical evolution of the performance of all caloric devices. A theoretical evaluation of the performance of residential caloric heat pumps composed of different caloric materials was conducted by Aprea et al.<sup>[148]</sup>

In the following section, we discuss the most significant research and accomplishments from the past five years, excluding minor improvements, and revisited studies from earlier work. **Figure 8** shows several common and layered principles of active magnetocaloric and electrocaloric regenerators. In addition to active caloric regenerators, **Figures 9** and **10** illustrate several heat transfer principles that have been employed in caloric and pyroelectric energy conversions.

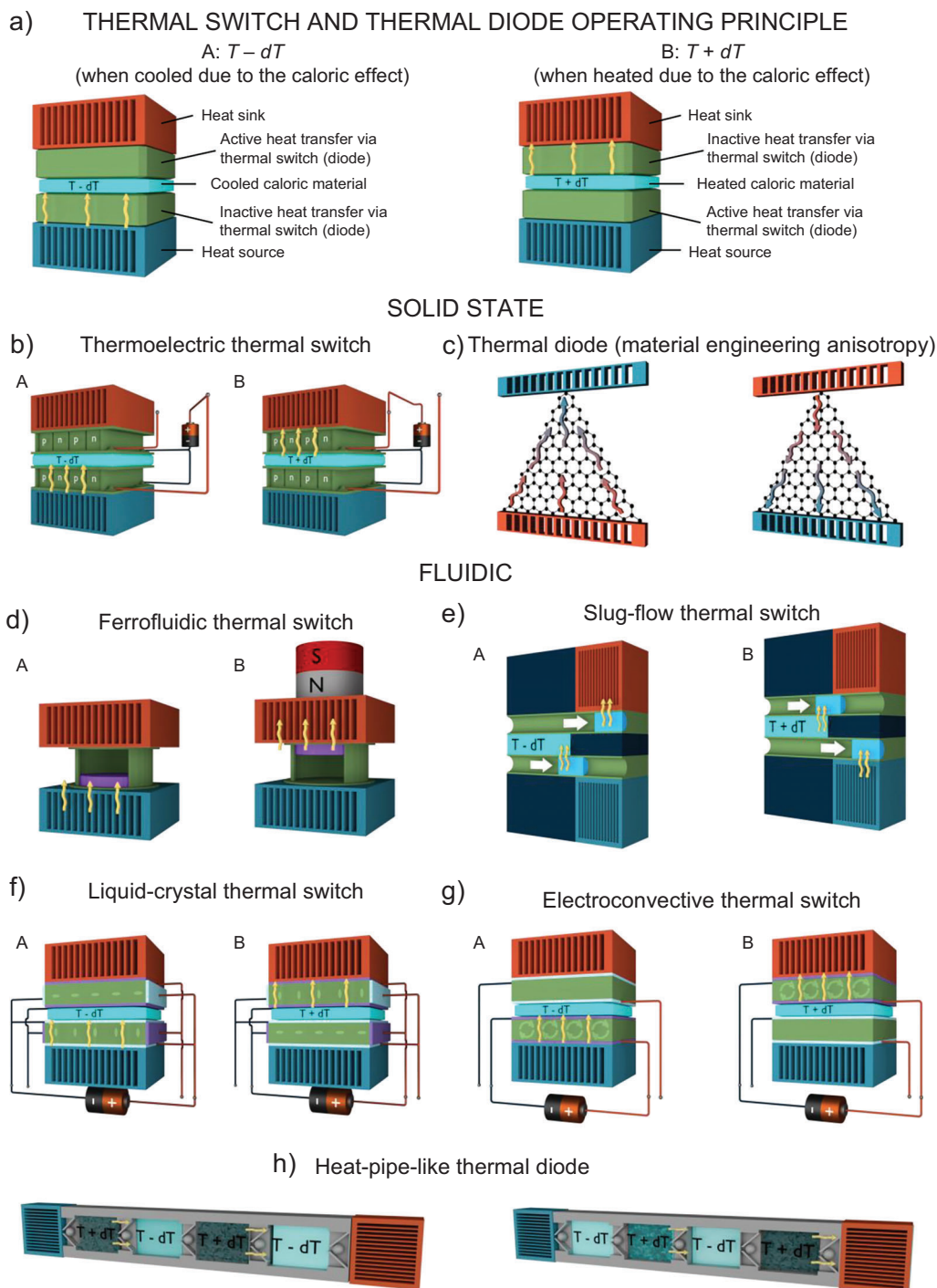
#### 4.1. Magnetocaloric Cooling and Heat Pumping

Major advances have been made in the field of cryogenics in recent years, with the exception of a few significant achievements in magnetocaloric cooling or heat pumping near room temper-

ature. Several recent studies have extended earlier work on the same prototype device by either improving it or applying it to new areas. **Figure 11** illustrates an example of the operation of a magnetocaloric cooling or heat pump device.

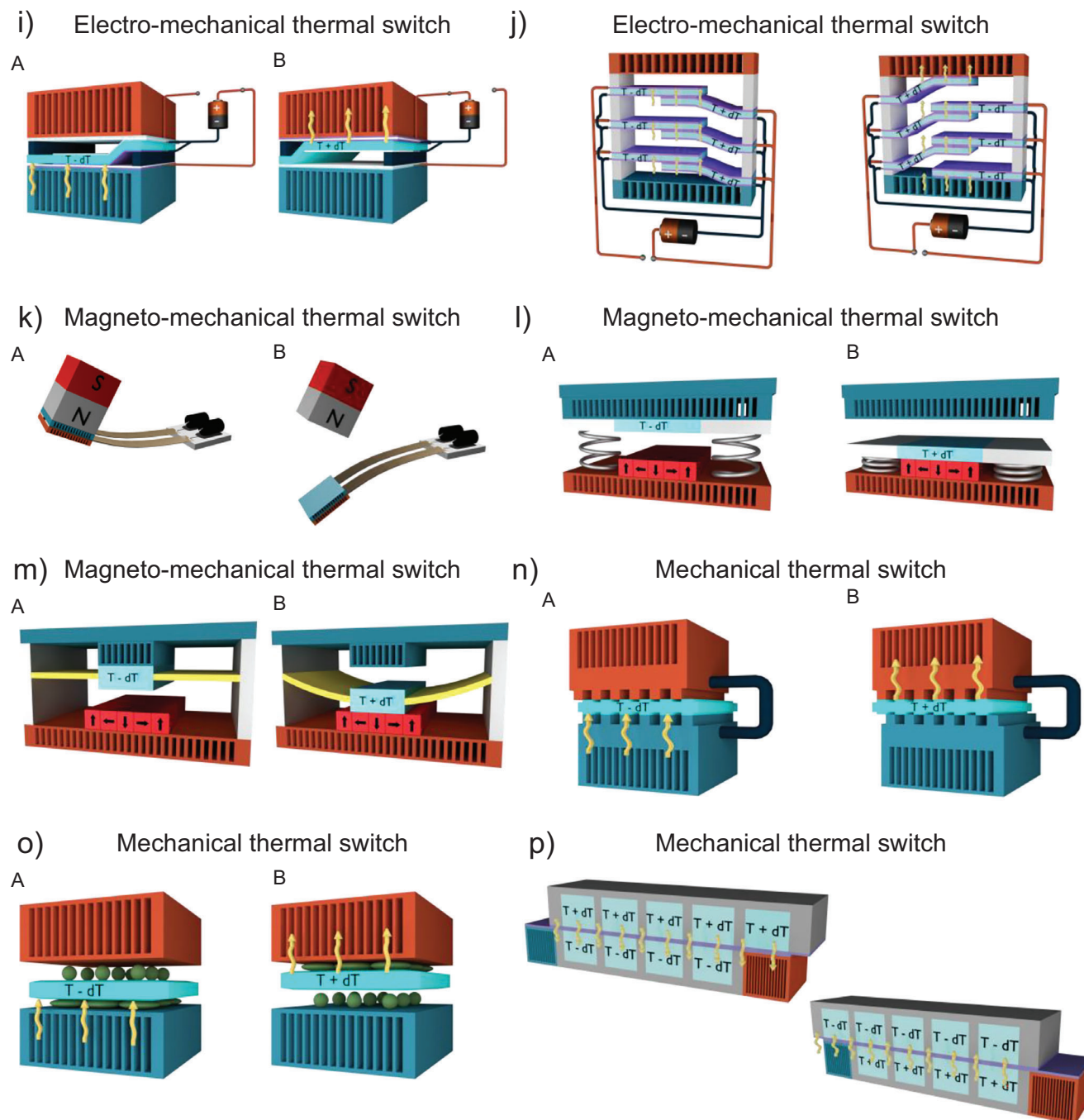
Most research on magnetic field sources still concerns the application of permanent magnets. Wieshou et al.<sup>[157]</sup> reported the optimization of rotating permanent magnet assemblies through topology and shape optimizations. In their study, Almeida et al.<sup>[158]</sup> revisited early research based on an unconventional approach to optimize magnetic field sources and the associated magnetocaloric effect by leveraging the demagnetizing effect. Another crucial area of research pertains to the processing and recycling of permanent magnets, with notable contributions from caloric researchers. For instance, the research conducted by Benke et al.<sup>[159]</sup> focused on the recycling of NdFeB permanent magnets. Additionally, Schäfer et al.<sup>[160]</sup> conducted research on specific alloys and offered a comprehensive overview of additive manufacturing methods for permanent magnets. In recent years, advancements in hybrid electro-permanent magnets have demonstrated the potential for static magnetocaloric devices to achieve energy efficiencies exceeding 70–80% when operated at frequencies exceeding 15 Hz. Furthermore, concerns regarding the potential overheating of electromagnetic structures have been disproved.<sup>[161,162]</sup> An interesting study was performed by Hirano et al.<sup>[163]</sup> who investigated the feasibility of applying high-temperature superconducting coils to a static magnetocaloric regenerator. The variation in the magnetic field was provided by the rotation of the magnetic shield over the regenerator.

In recent years, research on regenerators has encompassed various processing methods, such as additive manufacturing,<sup>[164,165]</sup> epoxy bonding of magnetocaloric granules,<sup>[166]</sup> and tape casting into ordered structures.<sup>[167]</sup> Additionally, studies have focused on evaluating the cyclic stability of first-order material regenerators.<sup>[44]</sup> Other research efforts have focused on the numerical or experimental



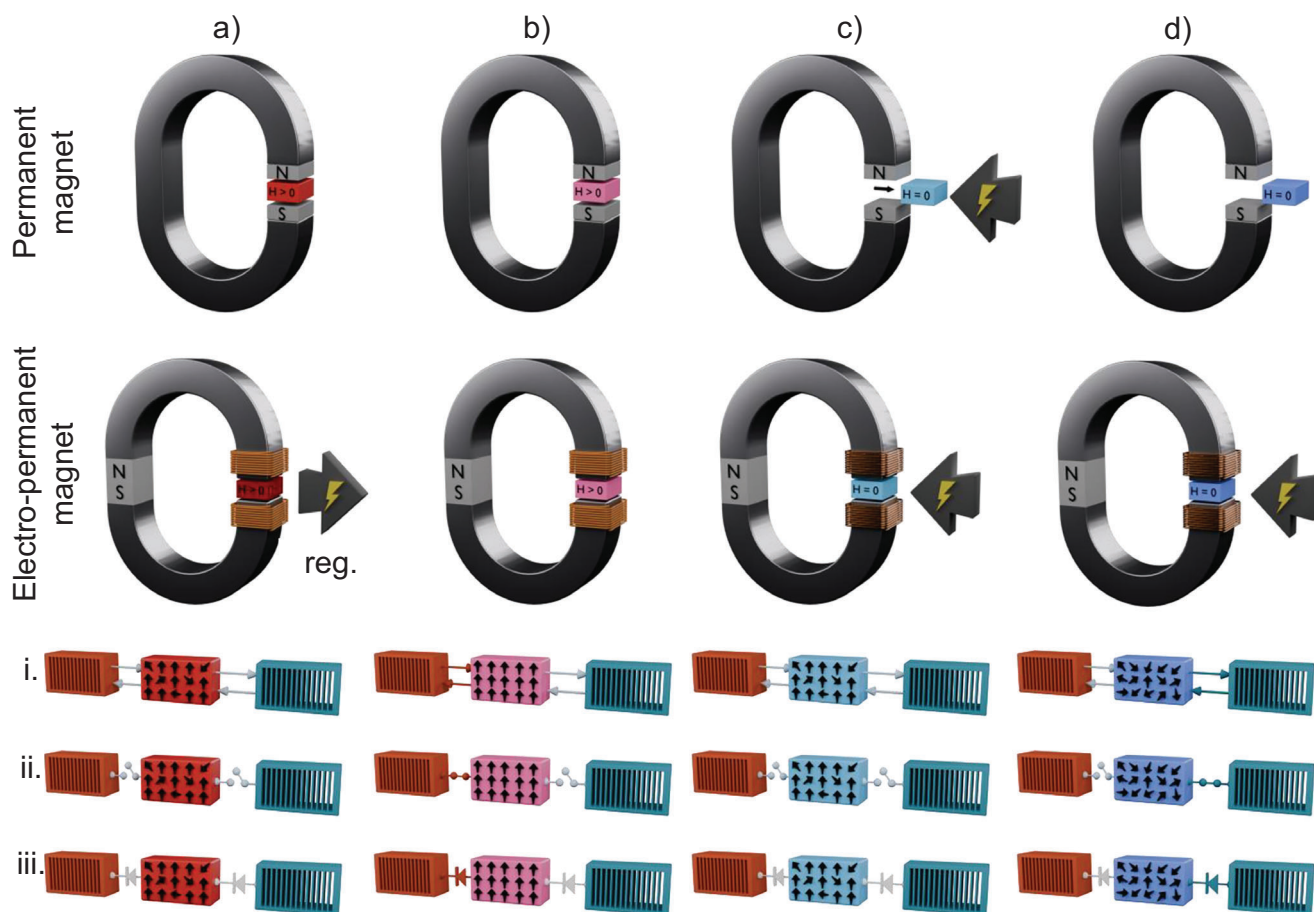
**Figure 9.** Different heat transfer principles using thermal switches and thermal diodes: a) basic operation of cooling and heat pump cycles when applying thermal diodes or thermal switches; b) thermoelectric switch, operating as the heat pump; c) solid-state thermal diode; d) response of a ferrofluidic thermal switch to a magnetic field by higher order particles, which can induce motion in the ferrofluid. Otherwise, when the full volume is occupied with the ferrofluid, the particles align with the magnetic field, thereby increasing the effective thermal conductivity; e) fluidic thermal switch, which causes the on–off operation of the thermal bridge via fluid; f) similar to ferrofluids, the liquid crystals respond in an on–off operation via ordered–disorder particle arrangements; g) electro- or magneto-convection occur when the gradient of the magnetic or electric fields and temperature are provided, leading to on–off operation; h) heat pipe thermal diode. Figures were reused with permission from the following sources: ref.[151] Figure 7b,d–g Reproduced from *Renewable Sustainable Energy Rev.* 2020, 118, 109 571, Thermal control elements for caloric energy conversion, K. Klinar, A. Kitanovski, Copyright (2020), with permission from Elsevier. ref.[152] Figure 7h reproduced under the terms of the CC-BY license,[152] 2020, Maier et al., published by Springer. ref.[153] Figure 7c reproduced under the terms of the CC-BY license,[153] 2020, Swoboda et al., published by Wiley-VCH Verlag GmbH & Co. KGaA. Please also refer to ref.[138]

## MECHANICAL THERMAL SWITCHES



**Figure 10.** Different heat transfer principles using mechanical thermal switches: i) single electrostatic mechanical thermal switch; j) multiple electrostatic mechanical thermal switches; k) resonant magneto-mechanical cantilever thermal switch; l) magneto-mechanical thermal switch with springs; m) magneto-mechanical elastic thermal switch; n) mechanical thermal switch; o) mechanical thermal switch with heat transfer grease or fluid; p) mechanical thermal switch with two caloric material beds. Figures were reused with permission from the following sources: ref.[151] Figure 8i,n,o reprinted from *Renewable Sustainable Energy Rev.* 2020, 118, 109571, Thermal control elements for caloric energy conversion, K. Klinar, A. Kitanovski, Copyright (2020), with permission from Elsevier. ref.[152] Figure 8h reproduced under the terms of the CC-BY license,<sup>[152]</sup> 2020, Maier et al., published by Springer. ref.[153] Figure 8l,m reproduced under the terms of the CC-BY license,<sup>[138]</sup> 2020, Kitanovski, published by Wiley-VCH Verlag GmbH & Co. KGaA. ref.[154] Figure 8k reproduced with permission, published by John Wiley and Sons, 2023. ref.[155] Figure 8j reproduced from A. Brađesko, D. Juričić, M. Santo Zarnik, B. Malič, Z. Kutnjak, T. Rojac, *Appl Phys Lett.* 2016, 109, with permission from AIP Publishing.<sup>[139]</sup> ref.[156] Figure 8p from Y. Wang, Z. Zhang, T. Usui, M. Benedict, S. Hirose, J. Lee, J. Kalb, D. Schwartz, *Science* (1979) 2020, 370, 129. Reproduced with permission from AAAS.<sup>[141]</sup>





**Figure 11.** a–d) Different processes involved in the exemplary Brayton-type refrigeration or heat device: constant zero field, magnetization, high constant magnetic field, and demagnetization; Permanent magnet: linear device with a permanent magnetic field source and the motion of the magnetocaloric material; Electro-permanent magnet: fully static operation with electro-permanent magnets. i–iii represents different principles of heat transfer: i) active magnetocaloric regenerator, ii) thermal switch, and iii) thermal diode. Arrows in the electro-permanent magnet indicate the work performed by the system or obtained from the system. The abbreviation reg. stands for regeneration.

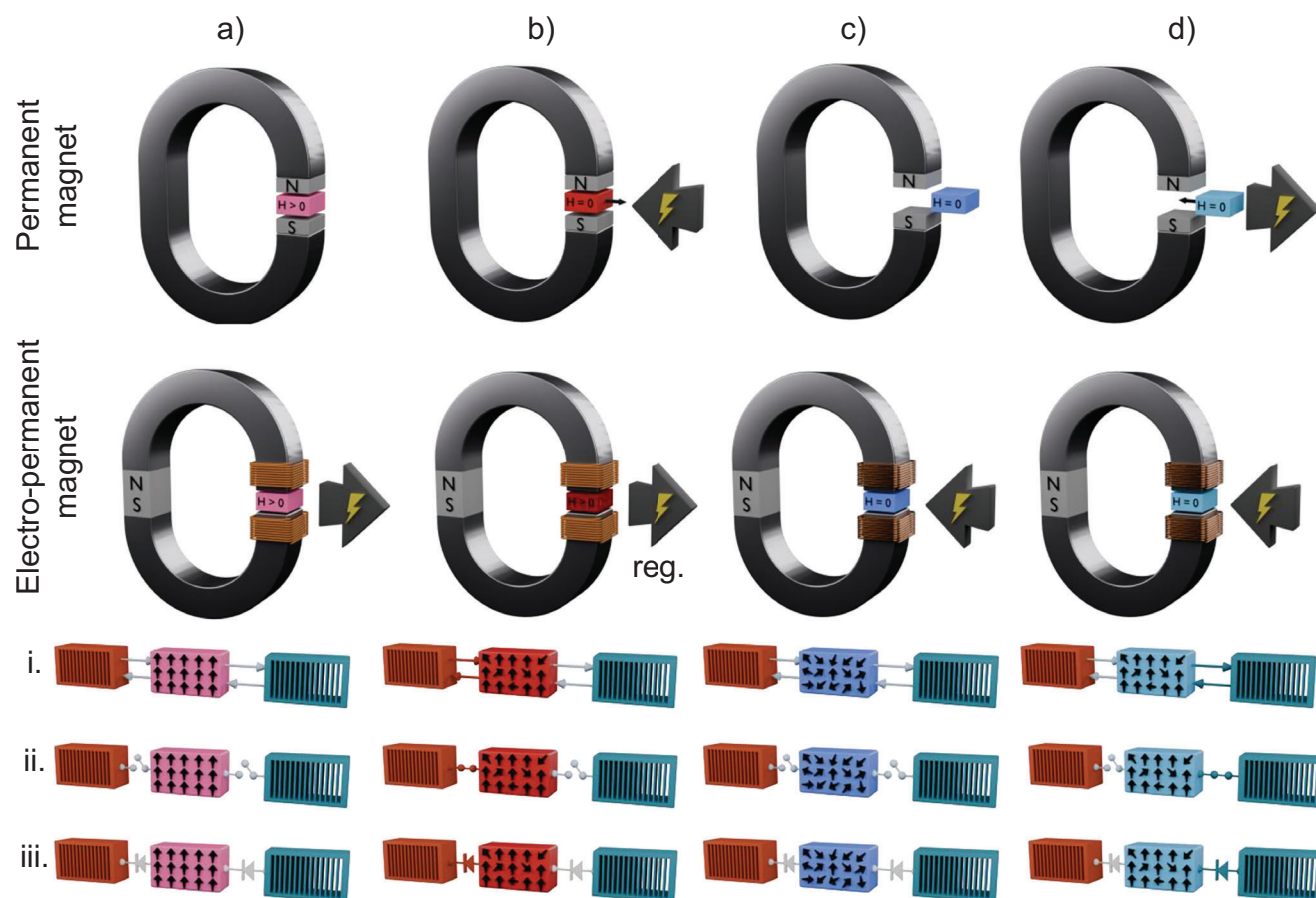
assessment of multilayered regenerators for applications near room temperature<sup>[168,169]</sup> and in cryogenics.<sup>[170,171]</sup> Notably, a significant contribution lies in the application of neural networks,<sup>[172]</sup> which are poised to emerge as the primary optimization method for multilayered regenerators across different caloric domains.

Thermal switches and thermal diodes have significant potential for enhancing the power density of caloric devices. Several numerical studies have been conducted in this area, including noteworthy research such as that of Silva et al.<sup>[173]</sup> who theoretically explored the essential characteristics of thermal switches for fluid-free applications. Petelin et al.<sup>[174,175]</sup> investigated an electrostatically actuated thermal switch capacitor. Klinar et al.<sup>[176]</sup> confirmed the significance of transient operation in solid-state thermal diodes through a theoretical analysis. Compared to steady-state operation, transient operation has the potential to significantly enhance the rectification factor. The use of heat pipes as thermal diodes in magnetocaloric devices was reported by Maier et al.<sup>[152]</sup> who observed significant improvements in the power density of a magnetocaloric device. However, generalized research on caloric devices incorporating thermal control ele-

ments, such as thermal switches and thermal diodes, is limited, with only two notable studies to date. In 2018, Silva et al.<sup>[177]</sup> introduced Heatrapy, an open-source Python framework tailored for transient 1D and 2D heat transfer simulations that is specifically applicable to modeling caloric devices with thermal switches. Another tool, TCCbuilder, stands out for its advanced features, including a material library and user-friendly graphical interface.<sup>[178]</sup> TCCbuilder is an open-source tool that streamlines the analysis of thermal control elements and circuits under both quasi-steady-state and transient conditions across a broad spectrum of applications. This tool aims to expedite the development of thermal management systems and enhance their power density, energy efficiency, reliability, and lifespan.

In recent years, research on magnetocaloric prototype devices has involved either the rotation of permanent magnet assemblies with static magnetocaloric regenerators<sup>[179]</sup> or the rotation of magnetocaloric regenerators within a static magnetic field source.<sup>[180]</sup> Some reported results highlight enhancements or testing of prior research devices,<sup>[181]</sup> specifically applied to areas such as larger-scale cooling devices,<sup>[182,183]</sup> wine coolers,<sup>[184,185]</sup> air conditioners,<sup>[186,187]</sup> and low-temperature heat





**Figure 12.** a–d) Different processes of the exemplary Brayton-type pyromagnetic device: constant zero field, magnetization, high constant magnetic field, and demagnetization; Permanent magnet: linear device with a permanent magnetic field source and the motion of the magnetocaloric material; Electro-permanent magnet: fully static operation with electro-permanent magnets. i–iii represent different principles of heat transfer: i) active pyromagnetic (magnetocaloric) regenerator, ii) thermal switch, and iii) thermal diode. Arrows in the electro-permanent magnet indicate the work performed by the system or obtained from the system. The abbreviation reg. stands for regeneration.

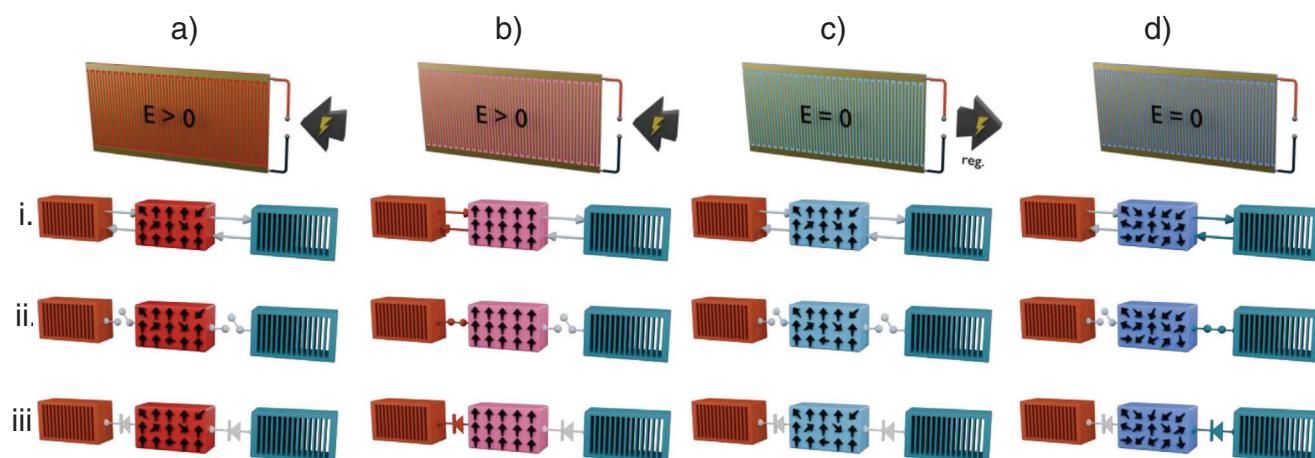
pumps.<sup>[188]</sup> Notably, Rajamani et al.<sup>[189,190]</sup> explored liquid metal and/or ferrofluid-based propulsion in relation to magnetocaloric devices. In addition, an intriguing and uncommon experimental study revisited the concept of sandwiching magnetocaloric materials between thermoelectric modules, with the latter serving as thermal switches.<sup>[191]</sup> As mentioned previously, cryogenics has been revisited within magnetocaloric systems, particularly focusing on theoretical studies related to the liquefaction of air,<sup>[192]</sup> hydrogen,<sup>[193–195]</sup> and liquefied natural gas.<sup>[196]</sup> It is important to highlight the significance of the experimental trials conducted by Hirano et al.<sup>[197]</sup> and Kamiya et al.<sup>[198]</sup> on hydrogen liquefaction, as well as the research by Archipley on methane liquefaction.<sup>[195]</sup>

## 4.2. Pyromagnetic Motors and Generators

In the field of thermomagnetism, the focus on regenerator design and related layers is currently limited compared to magnetocalorics. However, these components are poised to play a crucial role in enhancing efficiency across a range of operating temperatures and boosting power density. It is essential to dif-

ferentiate between thermomagnetic motors, which convert heat into mechanical work, and thermomagnetic generators, which directly convert heat into electricity, with or without mechanical motion. **Figure 12** illustrates an example of the operation of a pyromagnetic device.

The linear thermomagnetic motor evaluated by Kaneko et al.<sup>[199]</sup> utilized a double-C-shaped permanent magnet circuit, a packed bed Gd-based pistons, and a hydraulic circuit. The authors reported a specific power of 1.79 W per kg of Gd. In recent years, various examples of oscillatory motion devices have been reported, including an oscillating linear generator comprising a small hydraulic circuit, a permanent magnet assembly, a disc spring with two piezoelectric buzzers, and an attached  $\text{La}(\text{Fe},\text{Si})_{13}\text{H}$  plate.<sup>[200]</sup> Several other theoretical studies<sup>[201–204]</sup> and experimental evaluations<sup>[205]</sup> of linear thermomagnetic devices can be found in recent publications. Kohl et al.<sup>[206,201,202,204,203]</sup> explored a microscale thermomagnetic generator based on cantilever resonant oscillations, achieving an electrical power of 50 mW cm<sup>-2</sup> at a temperature change of 3 K. However, rotating thermomagnetic devices have not received much attention in recent years. In 2015, the SwissBlue company developed a 1 kW thermomagnetic motor.<sup>[207]</sup> However, further



**Figure 13.** a–d) Different processes of the exemplary Brayton-type refrigeration or heat devices: polarization, high constant electric field, depolarization, and constant zero electric field. (i–iii) represent different principles of heat transfer: active electrocaloric regenerator, thermal switch, and thermal diode, respectively. The arrows in panels (a–d) indicate the work performed by the system or obtained from the system. The abbreviation reg. stands for regeneration.

details on research and development, as well as recent activities, are scarce. Similarly, Hey et al.<sup>[208]</sup> investigated an experimental thermomagnetic motor using  $\text{Mn}_2\text{FeSn}$  as the thermomagnetic material, projecting a specific power of  $27 \text{ kW m}^{-3}$  with a maximum energy conversion efficiency of 2.1% post-optimization. Another study conducted by Mehmood et al.<sup>[209]</sup> reported a maximum power output of  $\approx 1 \text{ W}$  in their small device.

In contrast to solid working materials such as refrigerants, which are commonly used in thermomagnetic devices, ferrofluids offer an alternative approach for harnessing thermomagnetic effects. Noteworthy among related studies is the significant work of Chiolerio et al.<sup>[210]</sup> who developed a ferrofluid-based thermomagnetic generator in toroid form. They achieved a maximum extracted electrical power per unit of temperature difference of  $10.4 \mu\text{W K}^{-1}$ , with a second law efficiency approaching 20%.

### 4.3. Electrocaloric Cooling and Heat Pumping

Unlike magnetocaloric cooling, electrocaloric cryogenic cooling has not been the focus of significant research over the past two decades. Furthermore, there is no evidence of any prototype electrocaloric cryogenic devices being constructed. **Figure 13** shows an example of the operation of an electrocaloric device.

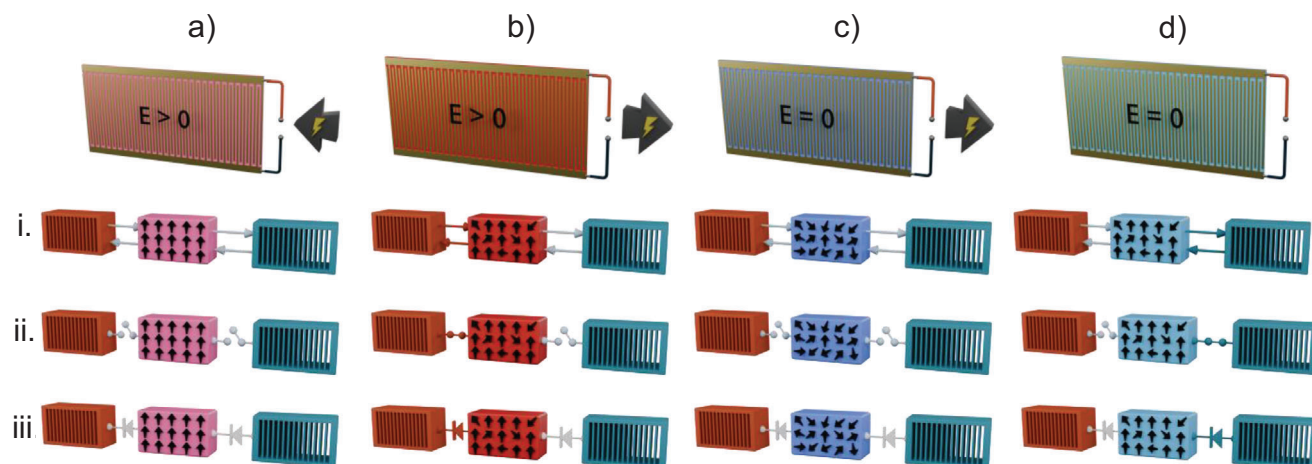
Unlike active magnetic regenerators in magnetocalorics, only a few examples of active electrocaloric regenerators have been reported in recent years. In this review, we present an interesting experimental study on a cascade electrocaloric cooling polymer tube<sup>[211]</sup> and a PST-based (lead scandium tantalate) parallel plate electrocaloric cooler, which achieved a maximum temperature span of 13 K and a maximum cooling power of  $12 \text{ W per kg}$  of PST multilayered structure.<sup>[119]</sup> Later, the same group reported a significant improvement by developing a double-loop electrocaloric heat pump and reaching a

maximum temperature span of 20.9 K with a specific cooling power of  $116 \text{ W per kg}$  of the PST multilayered structure.<sup>[118]</sup> Electric field recovery is crucial for energy-efficient operation of electrocaloric devices. In this context, Mönch et al.<sup>[212]</sup> reported a high efficiency (99.74%) for a charging/discharging converter.

Shi et al.<sup>[213]</sup> and Li et al.<sup>[214]</sup> theoretically assessed a unique cooler concept that revolves around the rotation of an assembly of electrocaloric regenerators, targeting the kW-level range for electrocaloric refrigeration systems.

Among the numerous theoretical and experimental research activities focused on electrocaloric coolers or heat pumps that incorporate thermal control devices, the work of Meng et al.<sup>[215]</sup> who integrated multiple units of electrocaloric polymer elements and an electrostatic actuation mechanism, is noteworthy. In this setup, every two adjacent elements operate in an antiphase to facilitate a continuous heat flow from the heat source to the heat sink. Their four-layer cascade device achieved a maximum temperature lift of 8.7 K under no-load conditions, with a maximum specific cooling power of  $78.5 \text{ mW cm}^{-2}$ . Bo et al.<sup>[216]</sup> conducted similar research, employing two actuating layers attached to a central processing unit (CPU) to achieve a maximum cooling effect of 23 K on the CPU.

Ma et al.<sup>[217]</sup> developed an interesting concept of a device that spans a few centimeters and operates without thermal control devices or working fluids. The device comprises three physically separated parts of multilayered electrocaloric ceramics based on doped  $\text{BaTiO}_3$ . The heat transfer processes in this system follow the Brayton cycle, relying on the physical motion of a specific part and the thermal contact with neighboring multilayered parts, heat sinks, or heat sources. Additionally, Wang et al.<sup>[156]</sup> reported another fully solid-state cascade system in which two sets of electrocaloric PST multilayer modules were thermally coupled and designed to move laterally relative to each other using an external actuator. The polarization electric field was switched synchronously, and the middle layer with



**Figure 14.** a–d) Different processes of the exemplary Brayton-type pyroelectric device: zero electric field, polarization, high constant magnetic field, and depolarization. (i–iii) represents different principles of heat transfer: i) active electrocaloric regenerator, ii) thermal switch, and iii) thermal diode.

anisotropic thermal conductivity enhanced performance. Operating at a low frequency (0.2 Hz), the device achieved a maximum temperature lift of 5.2 K under no-load conditions and a maximum heat flux of  $135 \text{ mW cm}^{-2}$  at a zero-temperature span. Other theoretical analyses have also highlighted the benefits of employing two independently controlled electrocaloric multilayers.

Further research on fully solid-state devices focuses on utilizing distributed thermal capacitances and resistances, in combination with spatio-temporally distributed electric field control, which has been shown to offer significant advantages in optimizing the temperature span of electrocaloric devices.<sup>[218]</sup>

#### 4.4. Pyroelectric Generators

This domain is the most rapidly developing solid-state energy harvesting technology. In general, we can divide pyroelectric generators into groups of nanoscale applications, which also consider sensors or small generators for powering electronics (i.e., power output from  $\mu\text{W}$  to  $\text{mW}$ ), mini-scale generators (i.e., power output from  $\text{mW}$  to  $\text{W}$ ), medium-scale generators (i.e., power output from  $\text{W}$  to  $\text{kW}$ ), and large-scale generators (above the  $\text{kW}$  range). For the last category, no studies or prototype devices have been reported. **Figure 14** illustrates an example of the operation of a pyroelectric device.

A special, rapidly developing domain concerns hybridization between different phenomena, such as triboelectric–pyroelectric,<sup>[219]</sup> pyroelectric–electrochemical,<sup>[220]</sup> thermoelectric–pyroelectric,<sup>[221,222]</sup> thermoelastic–pyroelectric,<sup>[223]</sup> thermomagnetic–pyroelectric,<sup>[224,225]</sup> piezoelectric–pyroelectric,<sup>[226–228]</sup> as well as combinations thereof.<sup>[229]</sup> Pyroelectricity represents the possibility of downscaling with rather simple and light structures, which makes the technology very interesting for remote and independent power supply for small-scale consumers, including different types of sensors, such as<sup>[230]</sup> infrared sensors and thermal imaging.<sup>[231–235]</sup>

temperature fluctuation sensors,<sup>[236]</sup> flexible proximity-sensing surfaces,<sup>[237]</sup> vacuum sensors,<sup>[238]</sup> bio-monitoring devices,<sup>[239]</sup> self-powered wireless sensor nodes,<sup>[240]</sup> and different types of wearable electronics.<sup>[241,242,242]</sup> Due to hybridization, which has been utilized in various studies, it is important to review studies on nanogenerators for sensors in emerging technologies<sup>[243–245]</sup> and for pyroelectric nanogenerators.<sup>[142]</sup>

Apart from sensorics, several different studies have focused on the use of waste heat sources for power generation. These include utilizing heat from electronics,<sup>[246]</sup> exhaust gases,<sup>[247]</sup> pyrolytic combustion of fuels,<sup>[248]</sup> the pyroelectric effect in catalysis,<sup>[249,250]</sup> thermal energy harvesting from friction generated in brake disks,<sup>[251,252]</sup> and heat generated by train vibrations.<sup>[253]</sup>

Other studies have concentrated on the energy conversion of solar energy, for which the temperature fluctuation is provided by different means and, in most cases, mechanical devices.<sup>[254]</sup> Researchers have focused on harvesting near-field radiation,<sup>[255]</sup> wearable solar–thermal–pyroelectric harvesters,<sup>[256]</sup> pyroelectric harvesters positioned on rotating windmill blades,<sup>[257]</sup> and pyroelectric harvesters with the inclusion of phase change materials,<sup>[258,259]</sup> among other approaches.

Surprisingly, because of the potential of this technology, very few research activities are concerned with medium-scale pyroelectric generators. From these results, it is worth mentioning the study by Lheritier et al.<sup>[260]</sup> who developed an energy harvester made of 42 g of lead scandium tantalate in the form of multilayer capacitors. The generator produced 11.2 J of electricity per thermodynamic cycle with a power density of  $4.43 \text{ J cm}^{-3}$ . The authors revealed a 40% Carnot efficiency for a 10 K temperature span. Another interesting theoretical study concerns hybridization, that is, the utilization of pyroelectric regenerators in a mechanical Stirling engine.<sup>[261]</sup> The results of the study showed that the pyroelectric regenerator increased the power output by 2.92%, producing 721 W compared to the 700 W produced in a conventional Stirling engine counterpart.



#### 4.5. Hybrid Devices Based on the Combination of Different Physical Principles

Notably, hybridization should not be confused with multicalorics, in which various physical phenomena are combined to amplify caloric effects. Currently, there is no evidence of a prototype demonstrator for multicalorics. In the coming years, we can expect a significant increase in the number of hybrid cooling devices, heat pumps, and power generation devices. These devices will combine various phenomena to overcome the limitations of a single technology. For example, the operating temperature availability of certain i-caloric or pyro-i materials can be improved by combining them with other materials and related phenomena or devices, not necessarily within the same group of i-caloric or pyro-i materials. Even the hybridization of i-caloric and pyro-i technologies with conventional vapor compression or thermoelectrics should not be overlooked.

A notable advancement in recent years is the hybrid triboelectric–thermomagnetic device developed by Rodrigues et al.<sup>[262]</sup> which achieved the highest voltage, current, and power density ( $\approx 67$  V, 2.5  $\mu$ A, and 17.9 mW m<sup>-2</sup>, respectively) under a constant temperature gradient of 41 K. Another noteworthy study by Zeeshan et al.<sup>[263]</sup> explored a hybrid piezoelectric–electromagnetic energy harvester integrated with a thermomagnetic engine.

The combination of an electrocaloric cooling/heating device powered by a triboelectric nanogenerator (TENG) was experimentally evaluated by Li et al.<sup>[264]</sup> who reported a 0.54 K temperature change achieved in 5.5 min of operation, with an energy conversion rate of 10.9% for the TENG with an area of 100 cm<sup>2</sup> operating at 3 Hz. A brief introduction to the triboelectric effect is provided in Section 6.2.2. Another interesting experimental study focused on the coupling of an electrocaloric device (based on electrostatic actuation) with a surface-integrated radiative heat sink/source-thermal regulation system. This hybrid device achieved a cooling power of 114 W m<sup>-2</sup> and a maximum temperature difference of 7.9 K.<sup>[265]</sup>

### 5. Challenges of Materials and Related Devices

The current power densities of all reported devices are significantly lower than that of competitive technologies. Furthermore, none of the studies have achieved notable success in attaining a high second law efficiency that theoretically exists. However, this may not be a major issue for sensors, where energy efficiency may not play a major role. A key issue prevalent in most studies is lack of innovation. Instead of addressing the major problems of i-caloric and pyro-i technologies, researchers often perform repetitive studies or incremental improvements when the solution requires a fundamental paradigm shift. Most researchers focus on application domains where it is evident that none of the studied i-caloric or pyro-i technologies can bring any competitive advantages to existing technologies in the long term, particularly concerning near-room applications of refrigeration, air-conditioning, and residential heat pumps.

The realistic carbon footprints of most i-caloric and pyro-i materials cannot compete with those of natural refrigerants. Additionally, issues concerning the use of rare earths, the toxicity of certain materials, and geographical dependence further hin-

der their competitiveness. Moreover, the power density (cooling power, heating power, or generated power per total mass of the system, not per refrigerant alone) of i-caloric and pyro-i devices is one to two, or even higher, orders of magnitude smaller than that of competitive technologies. Despite the well-established fact that conventional active i-caloric or pyro-i regenerative principles will not lead to a higher power density in devices, most researchers continue their studies in this direction. Furthermore, most studies on pyro-i power generation have disregarded heat regeneration, which is crucial for improved efficiency. Achieving a high or any field change is key to improving the second law efficiency of devices; however, this fact is primarily disregarded in pyro-i applications. Researchers often consider only the temperature dependence of magnetization (pyromagnetics) or polarization (pyroelectrics) and fail to recognize the importance of the applied field change when performing an efficient thermodynamic cycle. Magnetocaloric, pyromagnetic, electrocaloric, and pyroelectric technologies enable operation without moving parts, which is a key advantage and the primary reason for evaluating these technologies. However, numerous researchers still apply motional elements (e.g., motional magnetic fields, electric field sources or regenerators, and mechanical thermal switches with very high thermal contact resistance), which drastically affect efficiency.

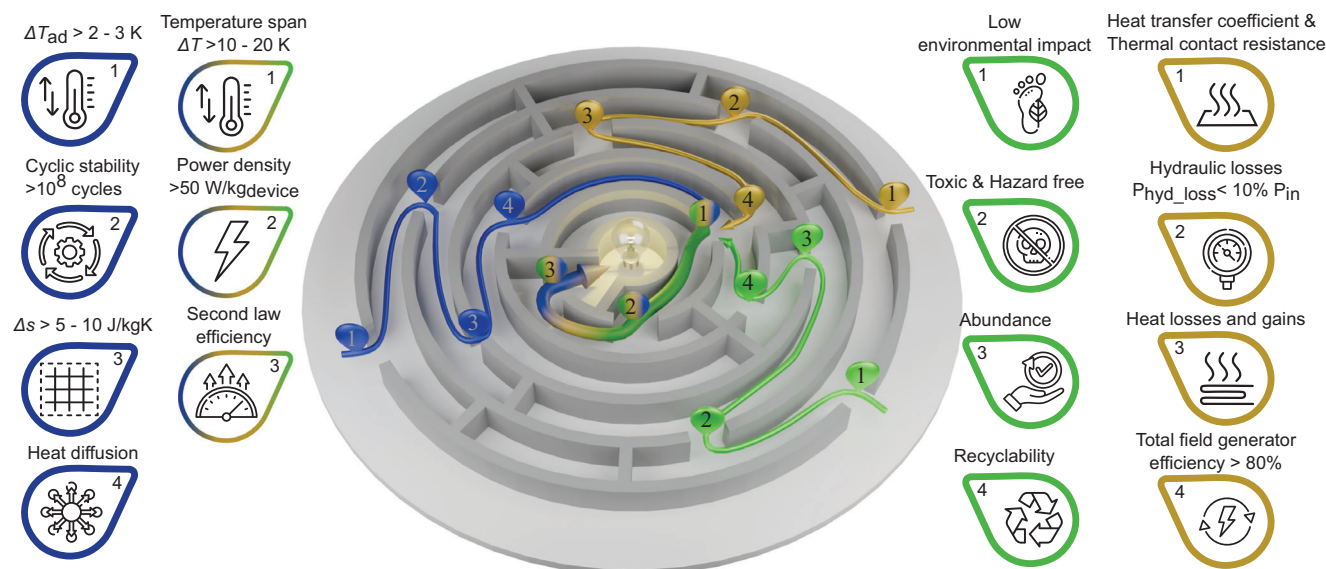
There is clear evidence of a large gap between basic and applied research. The number of publications on material studies is two to three orders of magnitude higher than that on applications. Concerning i-caloric and pyro-i materials, there are only incremental improvements in the related effects, particularly in the fields of magnetocalorics and pyromagnetics. In electrocalorics and pyroelectrics, the introduction of multilayered structures is a promising solution; however, there is a strong lack of research concerning processing methods to enable fast and highly efficient heat transfer. Many researchers have begun to explore solutions to the problem of extremely small i-caloric or pyro-i effects by studying multicaloric or multi-pyro effects. This approach can provide the advantage of a higher and more desirable overall adiabatic temperature change. However, in most cases, these solutions require the coupling of different field sources, which, again, increases the required mass of the field source and decreases the power density of the device. Studies on thermal control devices (thermal switches and thermal diodes), which enable the stationary operation of the evaluated technologies, are limited to a small number of engineering research groups globally. This is despite the fact that material researchers can provide crucial solutions to several related challenges. **Figure 15** shows the challenges associated with current devices and materials that must be overcome through future research and development.

Addressing the challenges described below (**Figure 15**) through innovation, advanced optimization strategies, targeting specific market niches, and transparent communication of results is essential to advance i-caloric and pyro-i technologies.

#### 5.1. Materials

From engineering and related thermodynamic perspectives, there is no practical distinction between i-caloric and pyro-i materials. Similar to refrigerants, these materials undergo different thermodynamic cycles, such as those observed in ORCs and





**Figure 15.** Challenges associated with current materials and related devices.

vapor compression. Therefore, the differentiation between i-caloric and pyro-i materials is irrelevant. Therefore, to create energy-efficient devices, materials with a high caloric effect in terms of adiabatic temperature changes are essential. In magnetocalorics and pyromagnetics, this is directly associated with the power density. In electrocalorics and pyroelectrics, the large adiabatic temperature changes required due to high electric fields push the limits of material breakdown, necessitating the development of significantly better multilayered structures beyond those available today. Any material and related device exhibiting less than 2–3 K of adiabatic temperature change cannot be a competitive technology. Note that the 2–3 K limit is related to the new solutions presented in this manuscript and not to state-of-the-art heat transfer solutions, for which the limit should be  $\approx 5$  K.

Furthermore, optimizing the layers of active materials in i-caloric or pyro-i devices to operate across a broader temperature span ( $>10$ – $20$  K) than the adiabatic temperature change is essential. This should not be performed by trial-and-error numerical simulations and experiments but should be optimized through the use of artificial intelligence and related methods. Another crucial feature concerns the change in isothermal entropy, which should be higher than  $5$ – $10$  J kg $^{-1}$  K $^{-1}$ . The change in isothermal entropy defines the specific energy capacity of a material. However, with an excessively low adiabatic temperature change, heat transfer cannot be accomplished within different processes (zeroth law of thermodynamics), even if the isothermal entropy change is infinite. Unfortunately, this has been disregarded in previous reports and publications. For conventional refrigerants and components of conventional refrigeration systems, heat pumps, and power generation devices, there must be a strict rule that all materials used in i-caloric and pyro-i devices must demonstrate abundance; have no or very low toxicity or handling hazards; exhibit low environmental impact over their total lifespan, from mining, production, and use; and have high levels of recyclability. All i-caloric and pyro-i materials and subsystems used in cooling, heat pumps, and power generation devices must

provide cyclic stability for a minimum of  $10^8$  cycles (for a device with a frequency of 10 Hz, this represents only 2780 h of continuous operation during its lifespan). This latter issue highlights the pressing need to find solutions that do not involve mechanical moving parts. It also defines the role of market niches and the related perspectives of a particular i-caloric or pyro-i technology. One of the strongly disregarded features of i-caloric materials is the dynamic caloric effect (i.e., the frequency-dependent caloric effect), which, except for magnetocaloric materials, has been disregarded in other domains of i-caloric materials.

## 5.2. Heat Transfer and Thermodynamics

The “classical” active caloric or pyro regenerative processes are energy-efficient only for very small power densities and operation frequencies in the range of 0.1 to  $\approx 3$  Hz, depending on the structure of the regenerator. Such a frequency in the “classical” active regenerative process can be usefully implemented only when a significant adiabatic temperature change is achieved, e.g., above 5 K (i.e., when using superconductors in magnetocalorics or pyromagnetics due to the insufficient magnetocaloric effect of existing materials, or when employing very fine structures of electrocaloric multilayers that can withstand strong electric fields). As a general rule for energy-efficient i-caloric or pyro-i devices, the heat transfer temperature difference between the i-caloric or pyro-i material and the fluid should be less than 10% of the adiabatic temperature change. For instance, if the required temperature difference for heat transfer of a certain capacity between the material and fluid is 1 K, this would require at least 10 K of adiabatic temperature change for efficient device operation. Therefore, it is crucial to provide a very high heat transfer area and/or high heat transfer coefficient. Achieving this requires mastering both the processing methods of materials and the thermal engineering of heat transfer and fluid dynamics.

In addition to the heat exchange between active and passive materials or fluids, the high-frequency operation of i-caloric and pyro-i devices requires fast diffusion through the i-caloric or pyro-i material. Researchers should not disregard this essential fact when tailoring or processing materials or when designing devices. The calculation of the diffusion time for a particular case requires simple methods, from which the simplest and most approximate value can be obtained by defining the dimensionless Fourier number as  $Fo = 1$ . Low energy efficiency at higher frequencies in classical active caloric or pyro regenerative processes is related to insufficient convective heat transfer, limited thermal diffusion through the material, and excessive viscous losses. These issues result in internal heat generation due to viscous dissipation and necessitate substantial pumping power, leading to challenges such as dead volumes, fluid maldistribution, process carry-over leakage, and losses in valves, among others. In general, to maintain high energy efficiency, viscous losses should not exceed 10% of input power. Attempts to apply energy-efficient thermal control devices (i.e., thermal switches or thermal diodes) have failed in existing studies owing to the significant problem of thermal contact resistance in mechanical or electromechanical thermal switches, including, in certain cases, too high thermal mass, too slow an actuation principle, too high energy consumption, or an inability to establish the regenerative process. Heat regeneration is imperative for caloric and pyro devices, although it may not be necessary for small sensors or micro power units. This has been best applied in magnetocaloric studies and surprisingly, to a significantly lesser extent in electrocalorics. Most known pyro-i devices do not apply heat regeneration, which is one of the reasons why these devices show very low second-law efficiencies. According to thermodynamic theory, achieving a high field change is crucial for maximizing energy efficiency. In pyro-i devices, simply tracing the magnetization or polarization curve against the temperature for a specific magnetic field is insufficient for creating an energy-efficient power generation device with high power density. The execution of the thermodynamic cycle is a pivotal step in the development of devices.

### 5.3. Field Sources

An inadequate magnetocaloric effect necessitates a very high magnetic field. This is directly associated with the very large mass of costly permanent magnet structures. The use of superconducting magnets is, on the other hand, limited to a particular application. To date, there is no evidence of the construction of a specially designed superconducting magnet system. Researchers have previously used commercially available systems that are not intended for refrigeration. Another problem, partially solved by the development of fully static devices implementing electro-permanent magnetic field sources, is related to mechanical moving parts and the necessity to optimize the recovery of any residual magnetic energy. Moreover, there is no solution that enables the static and high-frequency switching of high magnetic fields in superconducting magnets. Any electric field source lacking energy-efficient recovery during polarization/depolarization strongly limits and decreases the energy efficiency of electrocaloric coolers, heat pumps, or power generators.

### 5.4. Application Targeting and Devices

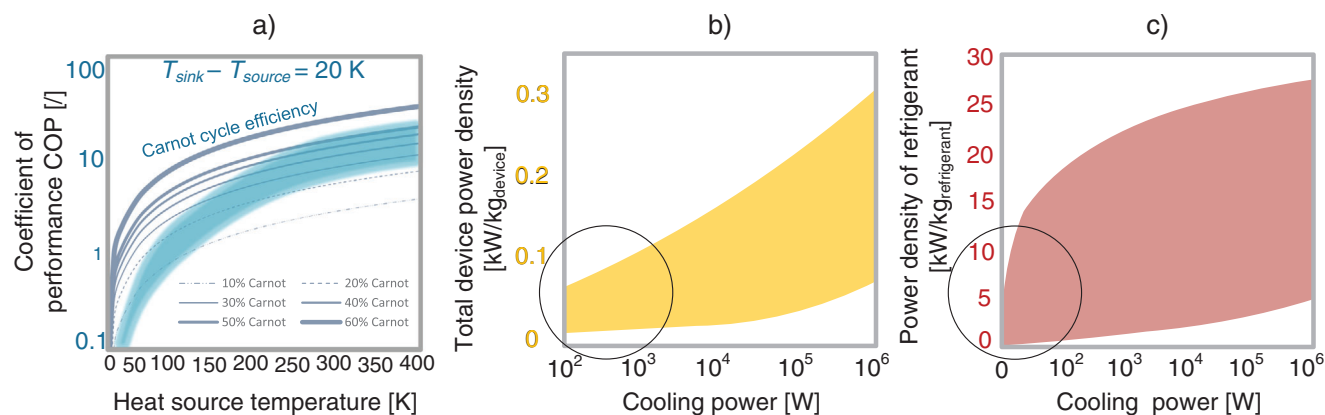
Targeting specific domains or market niches in which existing technologies have undergone long-term efficiency optimization, compactness improvements, and cost reductions presents a significant challenge. As mentioned earlier, this issue is especially relevant to near-room temperature heat pumps, cooling devices, and air-conditioners, as well as to devices with power generation over large temperature spans or operating at temperatures above 80 °C. Failure to address this issue can lead to difficulties in securing support from national and international funding sources. The aforementioned power densities of these devices are crucial for their market competitiveness. To bring i-caloric and pyro-i devices closer to actual market applications, their power density should be at least 50 W kg<sup>-1</sup> of the total mass of the device and for a minimum temperature span of 10 K, which includes all device elements for full operation. The devices should preferably possess no mechanical moving parts or related mechanisms because this causes issues related to irreversible losses and the lifespan of the device. Competitive refrigeration technologies are not as inefficient as those typically presented by certain researchers. The major figure of merit, in this case, is the exergy (Carnot or second law) efficiency of the total device, including its total consumption, for which researchers should focus on realistic solutions that can bring i-caloric or pyro-i technologies closer to those on the market. An example of this is shown in **Figure 16** in Section 6.

## 6. Roadmap for Future Materials and Applications

In this section, we propose potential solutions for the challenges identified in the evaluation of materials and related technologies. We also provide a roadmap that outlines these solutions within the timeline necessary for research until 2040, with a particular focus on under-researched domains. Furthermore, we conducted an online survey, inviting researchers from the caloric community belonging to institutes and industries to share their opinions on various questions. The survey link in ref. [266] remains open to participation in gathering broader opinions over time pertaining to specific problems and solutions in caloric and pyro technologies. The survey results are presented in the [Supporting Information](#).

### 6.1. The Future of Cooling, Heating, and Power Generation Materials

The abovementioned survey identifies material availability, cost of materials, and small adiabatic temperature changes as key limiting factors from a materials science perspective. The cost of raw materials and material availability are interlinked in the sense that the use of critical materials should be avoided to the greatest extent possible. A viable solution is available at certain temperature ranges, such as room temperature and high temperatures, where there are options that do not require critical materials. However, in the low-temperature range, which includes gas liquefaction, we remain highly dependent on rare earth-based materials. Unless breakthrough materials are developed, predicting abrupt changes is difficult.



**Figure 16.** An illustrative example of the conditions for heating and cooling, which researchers should consider when evaluating new types of devices: a) approximate trend of the COP and second law efficiency versus heat source temperature for existing cryogenic, cooling, and heat pump devices, derived from 303 different cooling/refrigeration devices from different manufacturers, as well as from references;<sup>[11,150,267–274]</sup> b) mass power density of vapor compression cooling devices (can also be considered for heat pumps), derived from data on 303 different cooling/refrigeration devices from different manufacturers; c) mass power density of refrigerants in vapor compression cooling devices (can also be considered for heat pumps), derived from data on 250 different cooling/refrigeration devices from different manufacturers. The black circles in (b) and (c) represent the most promising areas for caloric technology application.

A small change in the adiabatic temperature requires further investigation. Most studies focus on the maximum achievable value of temperature change, i.e., a “single shot” response, disregarding the influence of hysteresis. The optimal method of addressing this problem is to engineer materials to maximize the reversibility of the transition, such that the difference between the cyclic and single-shot adiabatic responses is minimal. This can be achieved by tuning the composition of the material toward a tricritical composition that separates the first- and second-order phase transitions, thereby minimizing hysteresis. This approach may have a cost associated with reducing the “single shot” adiabatic temperature change but would be advantageous if it leads to an increased cyclic response with respect to the starting (non-tricritical) compositions. Alternatively, tuning the kinetics of the phase transition would also benefit cyclability, and it is essential to identify the pathways that facilitate the transition from a metastable state to a more stable state. Statistical physics approaches related to “shortcuts to adiabaticity” will be instrumental in understanding the optimal paths to follow, and we anticipate significant advances in this area in the near future. Parallel to the approach of minimizing hysteresis, maximizing the single shot response through compositional tuning—assuming that the percentage of “wasted” temperature change during cycling remains consistent with composition changes—can also improve the situation. However, the search for new compositions may be limited unless novel strategies are developed.

With respect to these new strategies for identifying materials, high entropy alloys open a broad window of opportunity. As mentioned in Section 3, the recent demonstration of the feasibility of achieving a significant magnetocaloric response in HEAs, with the additional advantage of increased mechanical stability of the materials upon cycling, will certainly spur this line of research.

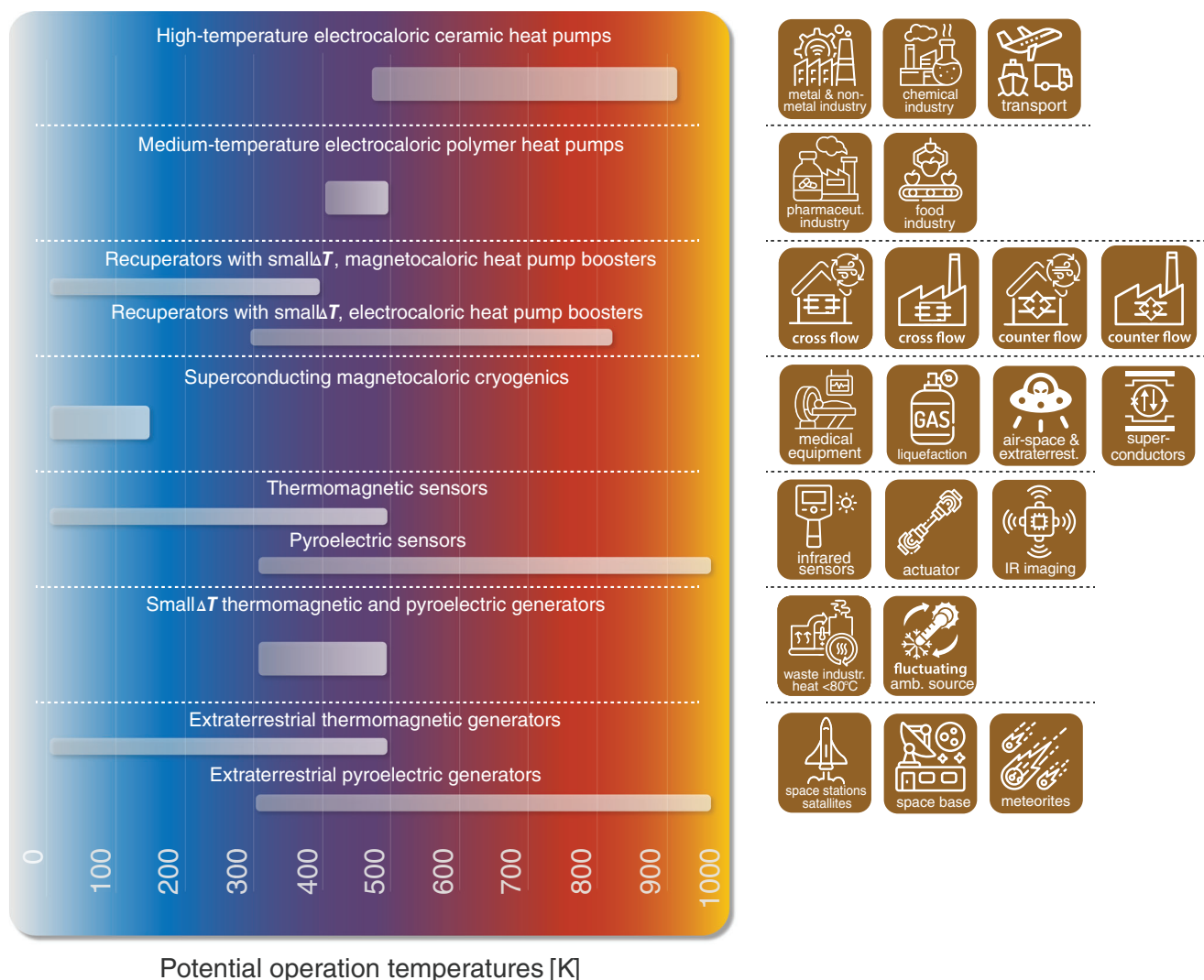
Another aspect that requires further research is the scarcity of materials at temperatures above 400 K. This is only minimally covered by amorphous alloys, HEAs, and, most significantly, metal monoborides. It is expected that MM<sub>2</sub>X and Heusler

alloys (including all-d-metal Heusler alloys), which can exhibit significantly high transition temperatures, can also be optimized for high-temperature heat pumping, a niche of applications with significant future potential.

Future applications of electrocaloric materials for heating and cooling, and pyroelectric materials for energy harvesting, lie in their ability to withstand higher electric fields. This involves enhancing the breakdown field and maintaining mechanical stability at elevated electric fields and temperatures during sustained cycling. Therefore, solutions that improve electrical and mechanical strengths are crucial. Multilayer capacitors with adiabatic temperature changes of less than 3 K are likely to fail to provide energy-efficient electrocaloric devices with good cooling densities. Similarly, multilayer capacitors with pyroelectric coefficients smaller than 50 mCm<sup>−2</sup>K<sup>−1</sup> are likely to fail to provide energy-efficient pyroelectric devices with good power densities. Moreover, similar to pyromagnetic materials driven using magnetic fields, pyroelectric materials driven by an electric field, with adiabatic temperature changes of less than 3 K, would not be suitable for practical harvesting devices. This is because energy-efficient harvesting thermodynamic cycles require active control of the temperature of the pyroelectric material using an electric field.

For any application, the maximum operating electric field should ideally not exceed 70–80% of the breakdown field. The downscaling of layers in multilayered structures and the application of new electrode topologies may prove essential for enhancing electrical and mechanical robustness. However, the scaling effects must first be better understood. The exploration of nanoscale and bulk metamaterial structures may also provide significant advantages.

Combining experimental and theoretical research is the key to improving the performance of materials. Theoretical tools of interest include finite-element modeling, first- and second-principle calculations, and the use of optimization algorithms and artificial intelligence methods, including neural networks. The experimental characterization of materials under different



**Figure 17.** Most promising applications are based on the temperature range of their utilization, which are expected to be developed by 2040.

ambient conditions and the unambiguous determination of key performance parameters remain challenging at small scales from the perspectives of reliability, repeatability, and accuracy.

Improving the performance of electrocaloric and pyroelectric oxides through B-site substitution in perovskites and strain engineering is another focus. Metal-free perovskites are also important owing to their lead-free nature, low environmental impact, and reduced health risks. Additionally, research efforts should focus on optimizing or manipulating the anisotropy of thermal properties in polymer electrocaloric materials to increase heat diffusion only in the desired directions, and new methods for printing and processing thin layers are needed. The environmental impact of polymer electrocaloric material production and disposal, including recycling, should be evaluated throughout the entire life cycle, considering the potential environmental impacts.

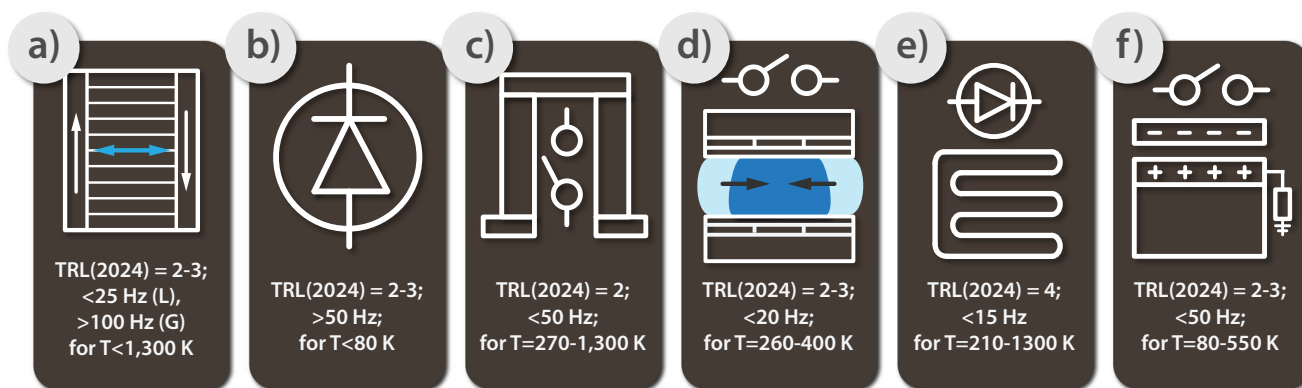
Furthermore, several aspects of electrocalorics and pyroelectrics remain largely unexplored. One such example is the exploration of low-temperature electrocaloric materials and

composites operating at sub-ambient temperatures, including cryogenic temperatures. Electrocaloric materials and devices can also provide good performance at high temperatures, making them central candidates not only for energy harvesting but also for medium- and high-temperature heat pumps operating from 400 to 1000 K. Another overlooked aspect is the research and development of electrocaloric fluids.

## 6.2. The Future of Cooling, Heating, and Power Generation Devices

Future research and development should focus first on these niches of applications, where there undoubtedly exists the potential for the superiority of the evaluated technology, which further requires good knowledge of all emerging and existing technologies per domain. The advantages of technology must be experimentally proven, or at least a clear solution to the challenges of





**Figure 18.** Six different heat transfer approaches for future research and development related to medium- and high-frequency devices: a) Hypereg<sup>[275]</sup> regenerator; b) solid-state thermal diodes; c) thin-film thermoelectric thermal switches; d) digital microfluidic thermal switches; e) heat pipe thermal diodes; and f) triboelectric thermal switches. TRL, technology readiness level.

reaching an advantage should be identified. The advantages to be considered are a high second law efficiency, which accounts for all losses; high power density per unit mass of a device or a system; use of recyclable, abundant, and environmentally friendly materials; longevity; and application in specific niches where no other competitive technologies exist or where the technology cannot be used. These advantages may be considered preconditions for future research.

Figure 16a displays the *COP* of state-of-the-art refrigeration and heat pump technologies in relation to the heat source temperature. This graph highlights the potential for improving energy efficiency by implementing advanced caloric cryogenic technologies. It underscores the importance of accurately reporting the power density per device and refrigerant used, as discussed earlier. While market-available technologies exhibit superior second-law efficiencies at near-room or higher temperatures, caloric heating and heat pumping can be effective for specific applications. Notably, and not shown in Figure 16, existing vapor compression and sorption heat pumps struggle to operate efficiently at temperatures exceeding 200–250 °C. This limitation opens up a new research domain for high-temperature solid-state heat pump technologies. Moreover, vapor compression cooling devices with low cooling power (below 50–100 W) achieve second-law efficiencies of  $\approx 20$ –35%, indicating room for improvement in vapor compression and potential for caloric technologies. However, for the latter, the primary challenge is to achieve not only sufficient efficiency but also adequate power density and cost-effectiveness, as illustrated in Figure 16b,c. The black circles in these figure panels represent the most promising areas for technological caloric applications. These aspects are not shown in Figure 16a. This omission is because, despite the high second-law efficiency of vapor compression systems at near-room or higher temperatures, there is a demand for small cooling and heat pump devices that operate with minimal temperature differences between their heat sink and heat source. These devices can serve as integral components of vapor compression and sorption systems and act as compact cooling/heating boosters or special heat recuperators with an integrated heat pump principle for cooling or heating purposes.

For power generation, existing ORC technologies offer good second-law efficiency across a range of input powers, start-

ing from a few kilowatts of output power, with the heat source temperature as low as  $\approx 80$  °C. However, for smaller devices and lower temperature levels below 80 °C, thermoelectric technology is currently the only available option; however, it is known for its low second law efficiency of  $\approx 10\%$  and problematic material types, such as Bi-Te, at these low-temperature levels. These issues highlight the potential for future developments in pyromagnetic and pyroelectric devices, which could offer improved efficiency and performance in output powers below 1 kW and temperature ranges below 80 °C.

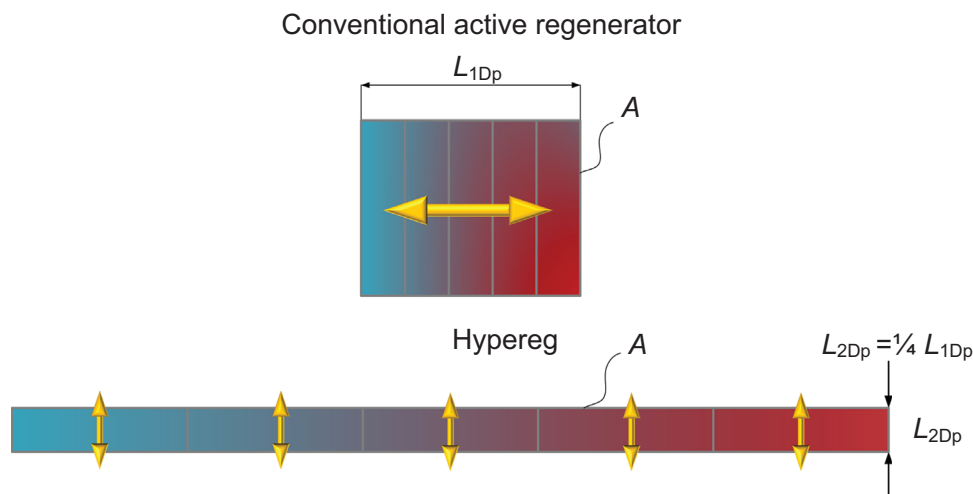
Second, we highlight the most critical applications that should be prioritized in future research activities (Figure 17). From this, we excluded all domains where vapor compression and sorption cooling are projected to remain unfeasible until at least 2040 due to the predicted lack of breakthrough solutions capable of overcoming the power density and second-law efficiency limitations of such devices. These domains can be addressed when the energy-efficient and high frequency ( $> 5$  Hz) heat transfer solutions are mastered, which are indicated in the text below (Figure 17).

#### Most Promising Applications Until 2040

##### a) Medium- and High-Temperature Electrocaloric Heat Pumps

In contrast to conventional magnetocaloric materials, electrocaloric materials offer a significant electrocaloric effect even at elevated temperatures. This characteristic is crucial because of their potential applications in domains where traditional technologies, excluding gas cycle-based heat pumps, such as reverse Brayton or Stirling, are unsuitable. Furthermore, gas cycle technologies, such as thermoacoustics, compete with calorics in the specific domain of heating temperatures above 200–250 °C. However, caloric devices are expected to deliver a higher power density and efficiency, and there is also potential for merging the two domains into hybrids.

Medium-temperature (e.g., 120–200 K) heat pumps are also significant because they can be downscaled for specific applications where vapor compression is inapplicable, and thermoelectrics are energy inefficient. Caloric heat pumps can advance in terms of temperature span; however, it is



**Figure 19.** Hydraulic difference between the conventional active caloric regenerator (above) and the Hypereg<sup>[275]</sup> regenerator (below). The pressure drop-related length and a ratio of  $\frac{1}{4}$  are depicted here as an example.

more likely that such applications will achieve temperature spans of  $\approx 30$  K (while exhibiting high energy efficiency and power density) over the next two decades. This is sufficient for certain applications involving melting, boiling, and steam production processes using waste heat recovery. However, for heating processes with fluctuating temperatures, these devices can recover waste heat from one process, increase the temperature, and utilize it in another process.

(b) *Heat Recuperative Low-Temperature Differential Magnetocaloric and Electrocaloric Heat Pumps*

First, heat recuperation does not necessarily require a large temperature span, which gives magnetocaloric (and electrocaloric) technologies an advantage. A heat pump operating as a heat recuperator not only transfers heat from one medium to another for heat recovery but also boosts the power and temperature levels to a certain extent. Such devices can be applied in air-conditioning (also in vehicles, where they presently do not exist) or various heat transfer processes involving gases or liquids, including small local boosters in hydronic cooling and heating systems. Conversely, heat recuperative boosters can be effectively applied at medium to very high temperatures using electrocaloric devices. In addition to space heating, cooling, or air-conditioning, such recuperators can serve various smaller applications in waste heat recovery.

(c) *Superconducting Heat Pumps for Heating and Cooling*

Superconducting magnets have been extensively studied for applications in magnetic energy storage, power generation, and other fields. Their energy consumption relative to the total energy output of the larger systems is minimal. Although there have been a few publications on the use of superconducting magnets for magnetocaloric cooling and heating, there are currently no prototype devices for this specific application, aside from cryogenic setups and tests of active magnetocaloric regenerators.

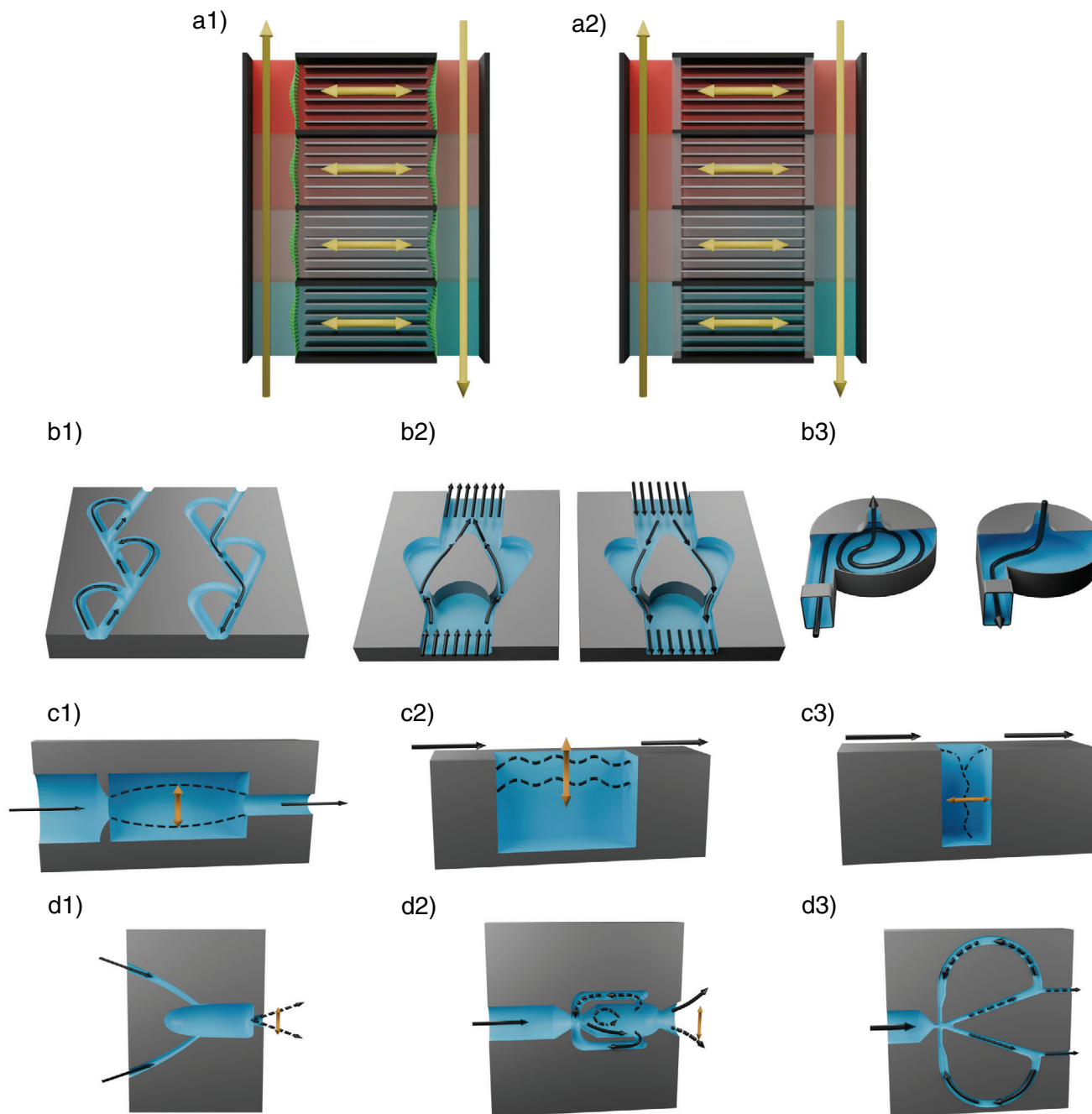
The development of a superconducting heat pump for cooling and heating requires a unique magnetic configuration or combination of magnets that can rapidly switch high electric currents on and off. Although switching off can occur within milliseconds, there is currently no solution for switching on. These devices should preferably be static, without the movement of the regenerator or magnetic shield within the magnetic field. The latter is questionable because it occupies valuable space in the magnetic field, whereas the former requires moving parts in the regenerator. In addition, there is no known solution for utilizing superconducting magnets to create magnetic field oscillations in the toroidal structure of regenerators. Despite these challenges, the utilization of superconducting heat pumps for cooling and heating represents a crucial direction for future research. This field has the potential to deliver powerful and energy-efficient solutions in the coming years.

(d) *Pyromagnetic and Pyroelectric Sensors*

Although pyroelectric sensors have already paved the way for potential future applications, there are also a variety of potential applications for pyromagnetic sensors or detectors. Unfortunately, no information is available on the utilization of the latter.

(e) *Low-Temperature Differential Power Generators or Motors*

These devices are not only intriguing for powering sensors but can also serve as a valuable energy source for remote, smaller consumers. The expected future power output is unlikely to surpass the kilowatt range. However, with the integration of energy storage and continuous operation, these devices can satisfy certain peak demands, thereby expanding their potential applications. The use of solar energy should not rely on active shading systems, as they significantly reduce irradiation energy, unless it is a natural or infrastructure-related source. Direct solar energy conversion into heat for power generation is energy-efficient only at high temperatures, such as in solar towers or concentrated solar systems, and not in low-temperature systems where most



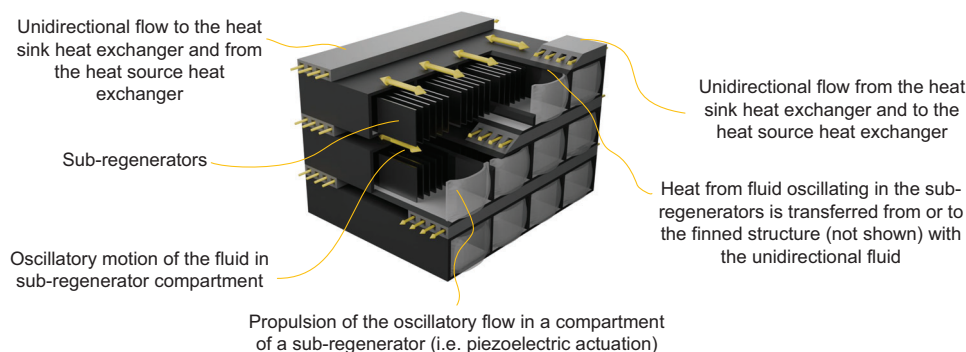
**Figure 20.** Novel high-frequency active caloric regenerator, termed “Hypereg”.<sup>[275]</sup> The in-parallel hydraulic configuration includes fluidic diodes: b1) Tesla valve; b2) a variation of the Tesla valve; b3) a radial fluidic diode.<sup>[276]</sup> Fluidic dynamic and resonant cavities: c1–c3).<sup>[277,278]</sup> Sweeping fluidic oscillators: d1–d3).<sup>[279]</sup>

of the exergy is lost. Thermal storage is essential for solar energy systems. Other promising systems include utilizing shallow geothermal energy through borehole heat exchangers, temperature differences between ambient air and water, temperature gradients in lakes or the sea, and waste heat recovery from various sources below 80 °C.

#### (f) Extraterrestrial Applications

All technologies under evaluation have significant potential for space exploration. They can function as sensors, sup-

ply power to satellites by leveraging the temperature differences between cooled electronics or space, or harness energy through temperature fluctuations, regardless of their temperature levels. In certain applications on moons or planets, these technologies may require two reservoirs for heat sources and heat sinks. Considering all magnetocaloric and electrocaloric materials, a vast array of energy-harvesting solutions can be explored between 0 and 1000 K. Another application involves heat pumps for cooling and heating at



**Figure 21.** A second configuration of the Hyperreg<sup>[275]</sup> regeneration principle.

temperatures that are uncommon on Earth. In these specific cases, caloric materials offer a substantial advantage over other systems as they are unaffected by gravity, low pressure, or extreme temperature levels. Currently, no research has been dedicated to the use of these technologies in space applications, which presents new opportunities for future research by the scientific community.

#### Future Heat Transfer and Regenerator Principles

We highlight the most important new approaches needed to achieve the development of this technology, with heat transfer being the most significant engineering hurdle. The heat transfer mechanisms illustrated in **Figure 18** are, in our opinion, the most promising, not only for high power density but also for high energy efficiency at related frequencies.

Among the six aforementioned heat transfer approaches, the Hyperreg and triboelectric thermal switches have not yet been discussed or revealed in the literature; therefore, we describe them in the following text.

To the best of our knowledge, for the first time, we unveil our patented concept<sup>[275]</sup> of the high-frequency (<25 Hz for liquids) active regeneration principle, the “Hyperreg.” The fundamental concept of the Hyperreg<sup>[275]</sup> regenerator, as depicted in **Figure 19**, involves reducing the pressure drop across the porous beds of a regenerator. This is achieved by configuring the fluid flow hydraulically in parallel rather than using the conventional series arrangement. The length, denoted by  $L_{\Delta p}$ , corresponds to the length required to calculate the pressure drop through the regenerator’s bed. Our initial assessment indicates that this principle marks a significant advancement in technologies that utilize regenerators to enhance power density while maintaining energy efficiency.

The principles depicted in **Figures 19–21** are based on the premise that in-parallel hydraulic connections (Hyperreg) offer significantly lower pressure drops than in-series connections (conventional active caloric regenerators). This design allows fluid to flow through the regenerators across their widths and not along their lengths. Two solution types are presented in this review: one in **Figure 20** and the other in **Figure 21**. In the first solution, two distinct configurations are outlined, both demonstrating counter-unidirectional fluid flow throughout the regenerator. This flow links the heat source and heat sink heat exchangers to the Hyperreg, with the fluid oscillating through the caloric

material sub-regenerators (**Figure 20a1,b1**). The controlled inner valves or porous membrane (green) shown in **Figure 20a1** facilitate appropriate flow oscillation through each bed. The alternative configuration in **Figure 20a2** is similar but incorporates motionless fluidic diodes or fluidic oscillators (gray areas at the entrances and exits of the sub-regenerator). These components enable fluid flow rectification and oscillation without moving parts. Various potential concepts are illustrated in **Figure 20b1–d3**.

In the specific instance of **Figure 21**, the working fluids in the sub-regenerators are hydraulically isolated from the unidirectional flow of the fluid along the regenerator. Consequently, heat is exchanged between the two fluids through an extended surface (not shown in the figure). This arrangement functions as a cross-flow heat exchanger, with one fluid circulating unidirectionally from the hot to the cold heat exchanger, and the second fluid oscillating in a direction perpendicular to the unidirectional fluid. The assembly in the figure comprises three layers, which can be adjusted based on the application by increasing or decreasing the number of layers and sub-regenerators. A greater number of sub-regenerators will result in a larger temperature span. Instead of employing a separate hydraulic propulsion device for each sub-regenerator, a single propulsion system can be used for multiple sub-regenerators.

The triboelectric thermal switch mechanism provides the advantage of adhesive forces between two surfaces of different materials, which is the most crucial factor related to thermal contact resistance in mechanical thermal switches. The triboelectric effect has mostly been investigated through mechanical oscillation energy harvesting in TENGs.<sup>[280,281]</sup> In this scenario, the contact between two different materials causes charge transfer between their surfaces. The material that gains charges becomes negatively charged, whereas the material that loses charges becomes positively charged; this phenomenon is also known as “contact electrification,” and the charge transfer can be attributed to electron, ion, and/or material movement between the two materials.<sup>[282]</sup> The inverse operation of the triboelectric generator, based on the applied charge, causes movement of the two materials, resulting in strong adhesion forces and, consequently, a relatively small thermal contact resistance. In this case, charge and discharge should be regenerated using electronic equipment. The inverse TENG principle can be effectively combined with electromagnetism. When further considering electric fields and mechanical movements, this principle



opens up a new area of future research for thermal switches applicable to all i-caloric and pyro-i technologies. This mechanism is currently under investigation at the University of Ljubljana.

## 7. Conclusion

The energy sector faces significant challenges, with 50% of final energy consumption attributed to heating and cooling and 50% of total global energy use resulting in valuable but unused waste heat. This highlights the critical importance of heating, cooling, and waste heat recovery in achieving decarbonization and sustainability goals. Despite the availability of market-ready and emerging cooling, heat pump, and heat harvesting technologies, caloric and pyrotechnologies remain essential in several domains of future energy conversion.

The development timeline for complex materials and technologies, such as caloric and pyro systems, is typically longer than that for less demanding products, regularly requiring 9–10 years to progress from technology readiness level 1 (TRL 1) to TRL 9. This extended timeline is often underestimated and should receive greater consideration from national and international research support programs.

To overcome these challenges and develop compact, energy-efficient devices, it is crucial to target specific niches where caloric or pyrotechnologies have distinct advantages over their competitors. Additionally, environmental concerns, such as material mining, processing, recycling, and carbon footprints, must be prioritized. Future studies should aim to achieve a high second-law efficiency and device power density while addressing these environmental concerns.

In this perspective paper, we propose solutions to the challenges faced by magnetocaloric, pyromagnetic, electrocaloric, and pyroelectric materials and devices. These recommendations aim to guide the research community in their future activities. We strongly encourage researchers to critically assess the technology, explore new solutions, and avoid replicating prior work to accelerate the development and market readiness of these technologies.

## Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

## Acknowledgements

A.K. and K.K. acknowledge the support of the Slovenian Research and Innovation Agency for the project TCCbuilder, an open-source simulation tool for thermal control circuits J7-3148, and research core funding no. P2-0223. V.F. and J.Y.L. acknowledge funding from PID2019-105720RB-I00/AEI/10.13039/501100011033; the Air Force Office of Scientific Research (FA8655-21-1-7044); the Clean Hydrogen Partnership and its members within the project HyLICAL (Grant No. 101101461); the European Innovation Council, funded by the European Union, through project CoCoMag (Grant No. 101099736); and the Research Council of Norway within project LIQUID-H.J.Y.L. acknowledges an EMERGIA 2021 fellowship from Junta de Andalucía (Ref. EMC21\_00418). X.M. acknowledges funding from the UK EPSRC (Grant No. EP/M003752/1), the ERC Starting Grant (Grant No. 680032), and the Royal Society.

## Conflict of Interest

The authors declare no conflict of interest.

## Keywords

cooling, electrocaloric, energy harvesting, heating, magnetocaloric, pyroelectric, pyromagnetic

Received: April 18, 2024  
Revised: August 14, 2024  
Published online: August 30, 2024

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